

PHYS 4371/6371 LECTURE, UNIT 2

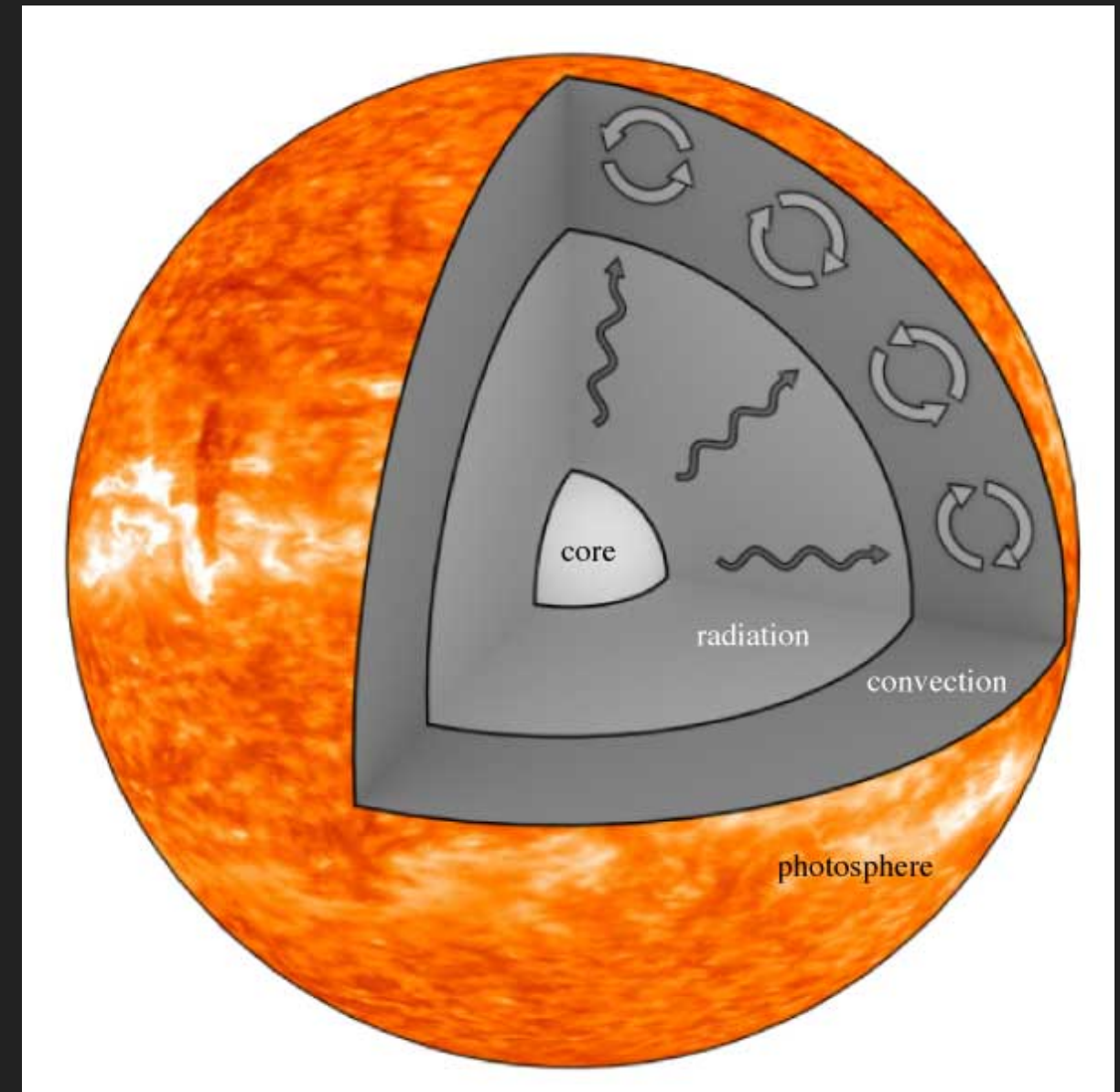
EQUATIONS OF STELLAR STRUCTURE

STELLAR ASTROPHYSICS (P. 2.1-2.8; C9.1,10.1; K2-5)

R. Kehoe - Stellar Astrophysics, 10/21

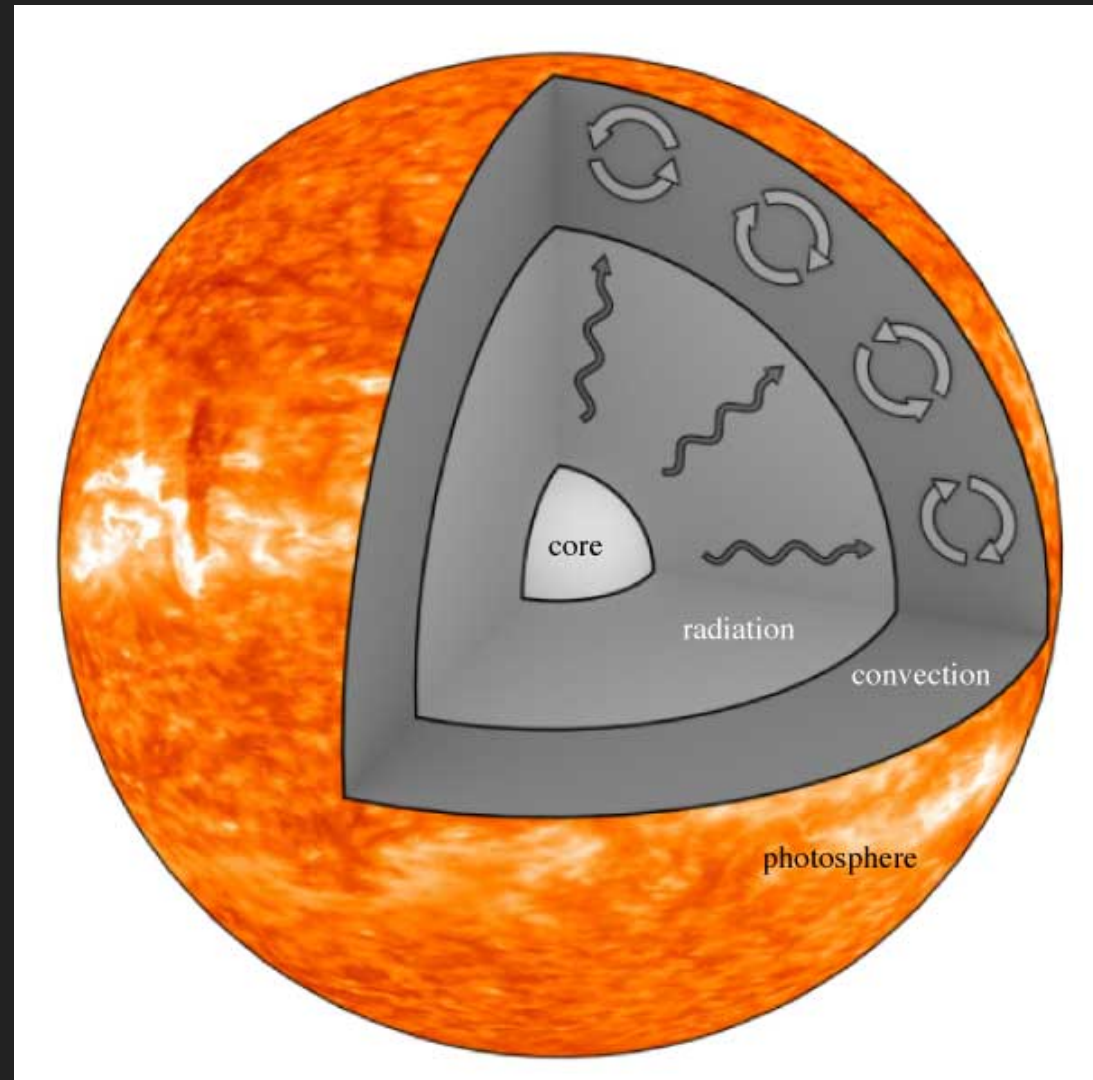
HOW DO WE UNDERSTAND STARS?

- ▶ Why does HR Diagram look like it does?
 - ▶ Why are some areas sparsely/densely populated?
 - ▶ Why are some contours shaped as they are?
- ▶ What is the structure of a star?
- ▶ What are the timescales of stellar life?
 - ▶ Are there different periods?
 - ▶ How are stars born?



INITIAL IDEALIZATIONS

- ▶ What forces impacting stellar structure?
 - ▶ Bound by self-gravity
 - ▶ Gravitational potential energy
 - ▶ Radiates energy supported by internal source(s)
 - ▶ Potentially more than one source
- ▶ Static – properties unchanging in time
- ▶ Spherical geometry
 - ▶ Ignore rotation
- ▶ Isolation
 - ▶ Exclude binary, multiple systems
 - ▶ Closest star $> 10^7 R_{\odot}$
 - ▶ Gravity and radiation diminished by next star $\sim 10^{-15}$
- ▶ Uniform initial composition
 - ▶ Very similar for many stars, interstellar medium (ISM)
 - ▶ Metal abundances < 0.001 to a few %
- ▶ No mass loss: Mass loss from fusion reactions, $E = mc^2 \ll$ total mass



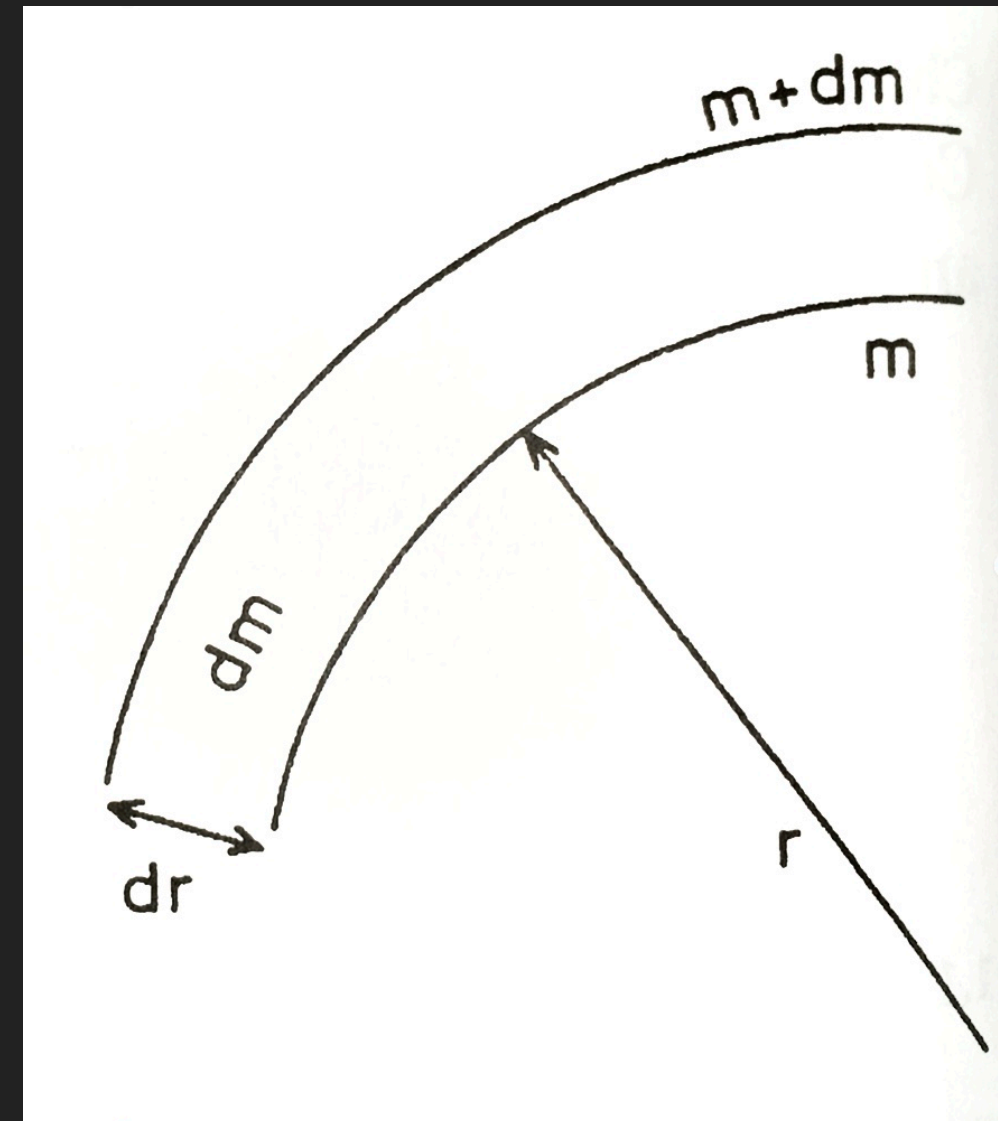
75% H by weight

25% He

Some O, C, N

CONSERVATION OF MASS

- ▶ Consider a star with
 - ▶ Mass, M
 - ▶ Photospheric radius, R
 - ▶ Photon luminosity, L
 - ▶ And some radial-dependent density, $\rho(r)$
- ▶ If above true and spherically symmetric
 - ▶ Properties depend only on radius, r
 - ▶ $M(r)$: enclosed mass out to radius ' r '
 - ▶ $L(r)$: enclosed luminosity
 - ▶ $P(r)$: pressure at ' r '
 - ▶ $T(r)$: temperature at ' r '



EQUATION OF CONSERVATION OF MASS

- ▶ A thin, spherical shell, thickness dr

- ▶ If $dr \ll r$, then $dV = 4\pi r^2 dr$

- ▶ Thin shell mass would be: $dM = 4\pi\rho r^2 dr$

- ▶ This can be written as

$$\frac{dM}{dr} = 4\pi\rho r^2 \quad \text{1st Equation of Stellar Structure (P1.5)}$$

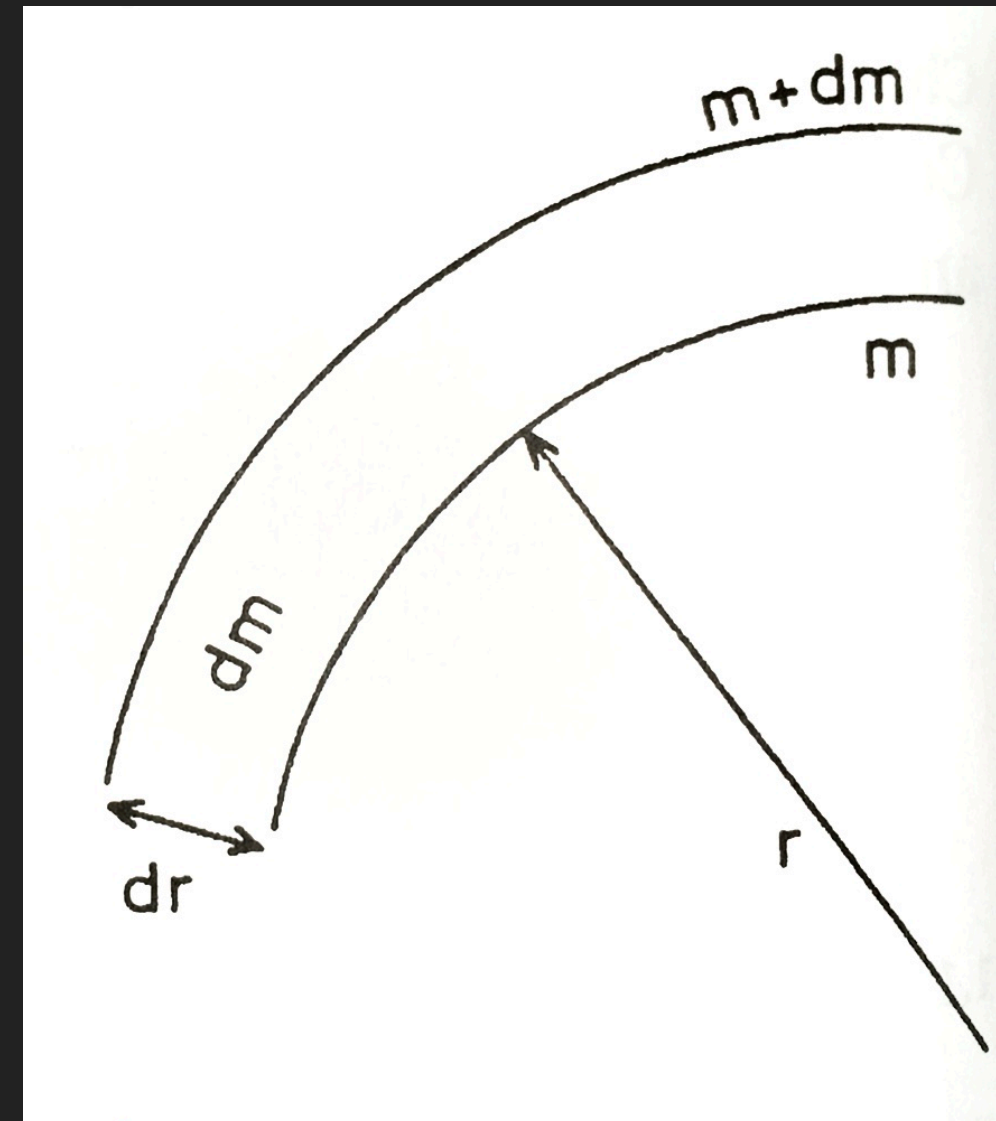
- ▶ Sometimes useful to write in terms of enclosed mass

- ▶ As radius of shell changes, amount of mass constant

$$\frac{dr}{dM} = \frac{1}{4\pi\rho r^2}$$

- ▶ Enclosed mass out to radius, r

$$M(r) = \int_0^r 4\pi r^2 \rho(r) dr$$



DYNAMICAL TIMESCALE

- ▶ Gravitational potential

$$\Phi(r) = -\frac{GM}{r}$$

- ▶ Acceleration is $\ddot{r} = -\frac{GM}{r^2}$

- ▶ Using Kepler's 3rd law

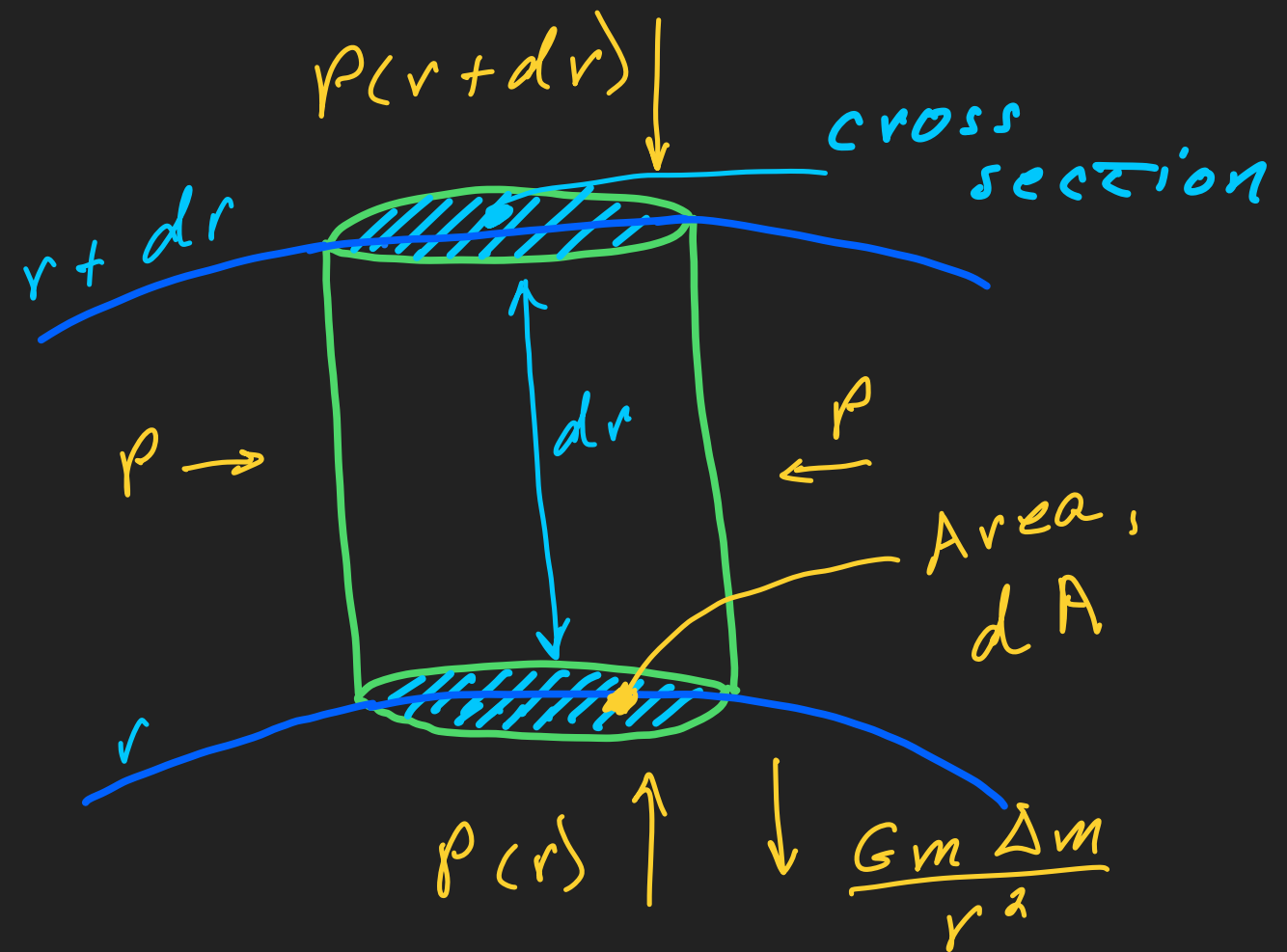
$$t_{ff} = \left(\frac{\pi^2 R^3}{8GM} \right)^{1/2} = \sqrt{\frac{3\pi}{32G\rho}}$$

Free Fall Time (~P2.57)

- ▶ Sun's mean density is 1.41 g/cm^3
 - ▶ So $t_{ff} \sim 1800s$
- ▶ The Sun is _much_ older than a half hour!
- ▶ Must be outward force to balance gravity
 - ▶ Since the Sun's age of $\sim 4.6 \text{ Gyr} \sim 10^{14} \times t_{ff}$
 - ▶ Gravity and outward pressure balance to 1 part in 10^{14} !

OPPOSING FORCES

- ▶ Stars are made of matter
- ▶ Consider cylinder of mass dm
 - ▶ Base is ' r ' from star center
 - ▶ Axis length, dr
 - ▶ Cross sectional area, dA
 - ▶ ~uniform density, ρ
- ▶ Two opposing forces
 - ▶ There must be some pressure, P , that is balancing gravity
 - ▶ **Hydrostatic equilibrium**
 - ▶ Both are directed radially
 - ▶ Due to spherical symmetry



BALANCING PRESSURES

► Pressure

- Exerted by gas 'above' and 'below' material in cylinder
- Forces on sides cancel
- Force on this section from pressure:

- Top minus bottom

$$F_p = PdA - (P + dP)dA = -dPdA$$

- Outward pressure, so remove -ve sign

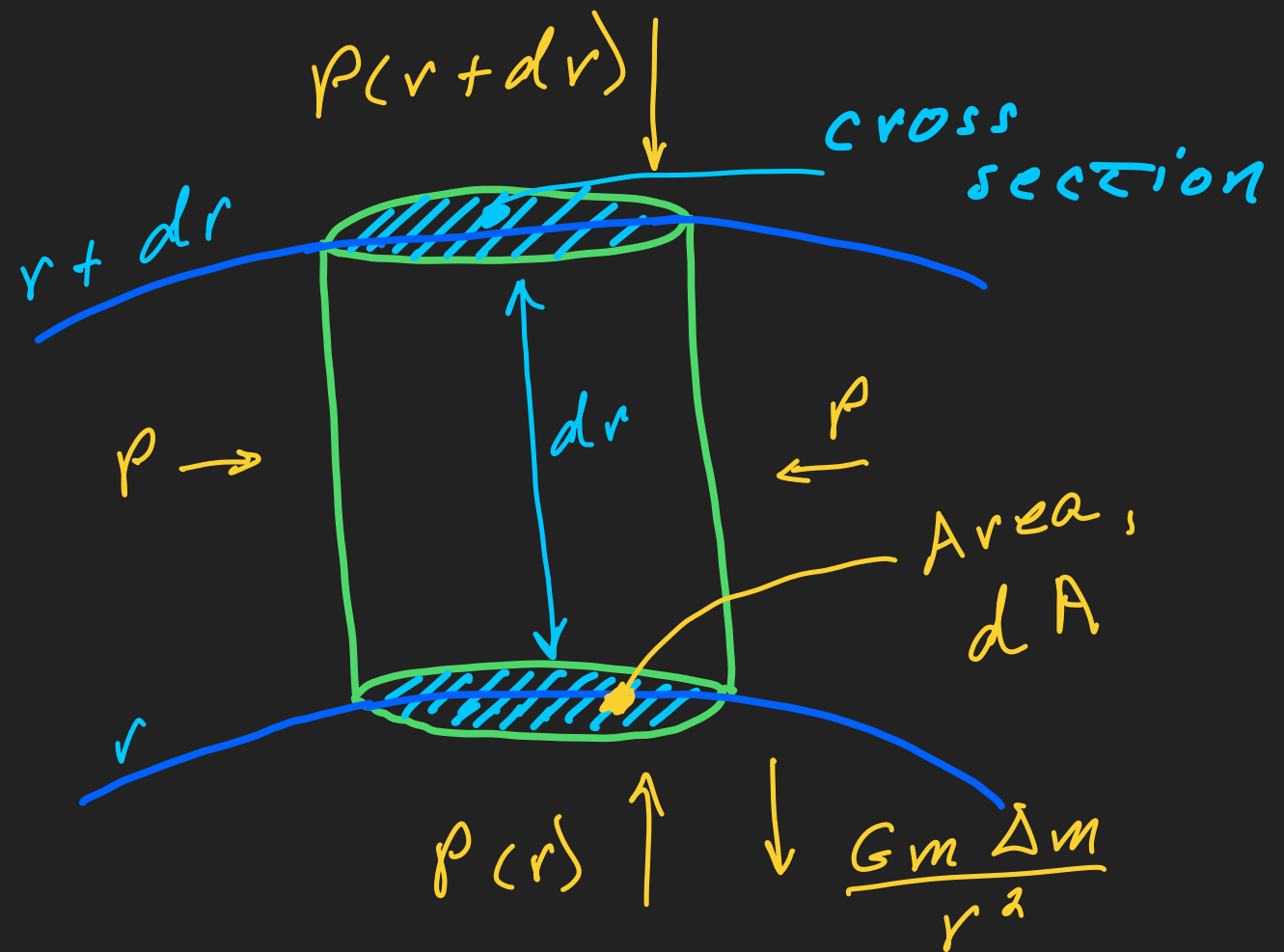
► Gravitational force

- From mass of sphere interior to radius, r
- Total mass of the cylinder:

$$\rho(r)drdA$$

- The gravitational force on the cylinder:

$$F_g = -\frac{GM}{r^2}\rho drdA$$



EQUATION OF HYDROSTATIC EQUILIBRIUM

- ▶ The condition for perfect balance
 - ▶ No net force, or acceleration, on cylinder
- ▶ Setting the forces equal to each other

$$dP dA = - \frac{GM}{r^2} \rho dr dA$$
$$\frac{dP}{dr} = - \frac{GM}{r^2} \rho$$

2nd Equation of
Stellar Structure

- ▶ Writing in terms of enclosed mass

$$\frac{dP}{dM} = \frac{dP}{dr} \frac{dr}{dM} = - \frac{GM}{4\pi r^4}$$

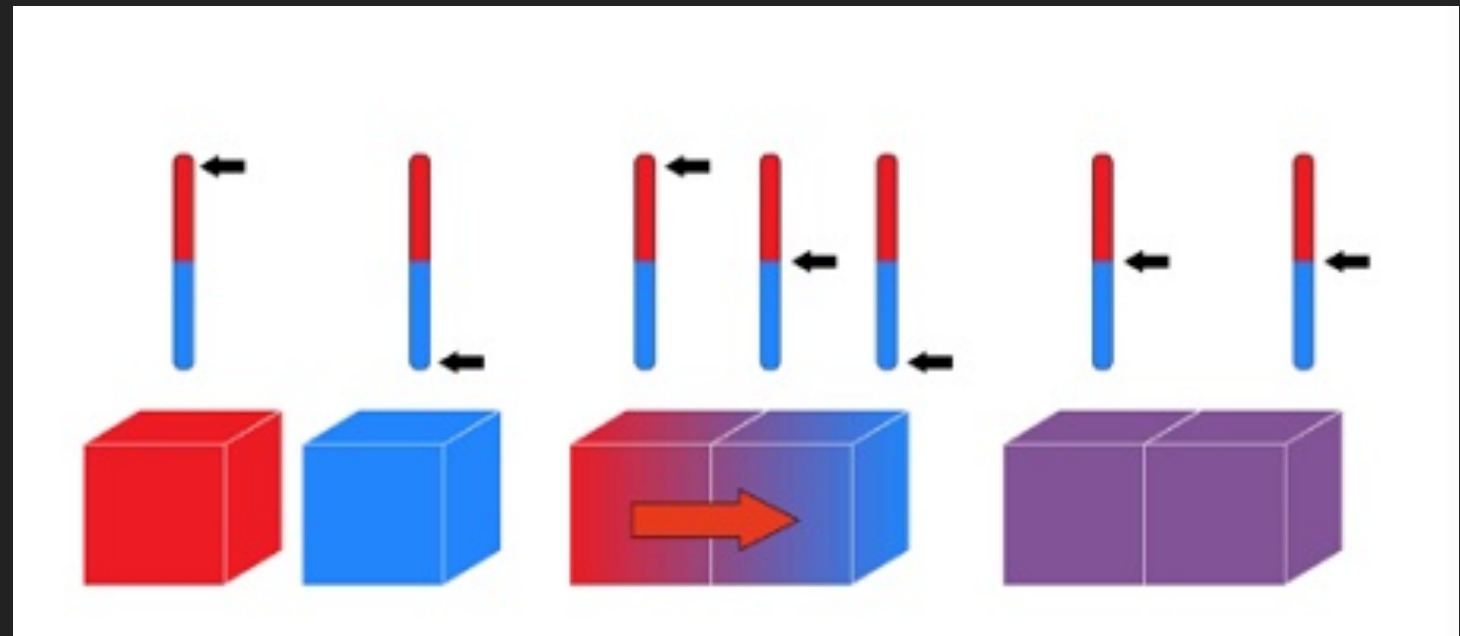
- ▶ What does it tell us?
 - ▶ Need a pressure gradient to counteract gravity
 - ▶ Since gradient always negative
 - ▶ Pressure must decrease with increasing radius

DISCUSSION QUESTION

Calculate the rate of change of pressure in the Sun at a radius of $0.9R_{\odot}$. Assume a density of 1 g/cm^3 , and a radius of $695,000 \text{ km}$. The total mass of the sun is approximately $2 \times 10^{30} \text{ kg}$.

THERMAL EQUILIBRIUM

- ▶ Stars emit radiation, i.e. lose energy
 - ▶ But stay the same brightness
 - ▶ Must not be cooling off
- ▶ Flow of heat (thermal energy)
 - ▶ Always in direction of decreasing temperature
- ▶ Star in 'thermal equilibrium'
 - ▶ Same amount of energy coming into a region (eg. Photosphere) as radiating away
 - ▶ Requires generation of new energy



EQUATION OF THERMAL EQUILIBRIUM

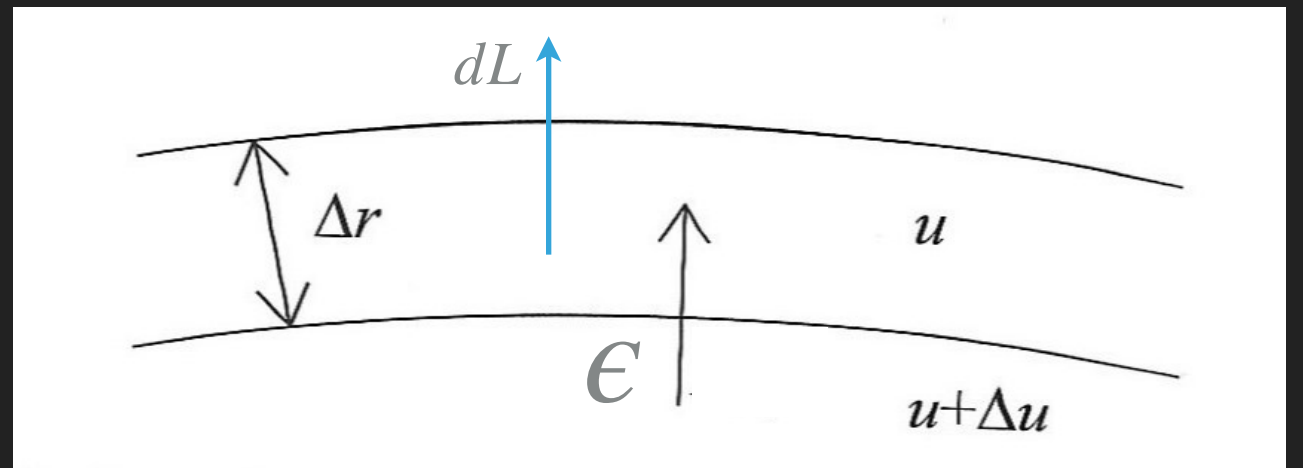
- ▶ Consider a thin, spherical shell of star
- ▶ Temperature constant if $dL = \epsilon dM$
 - ▶ dL is lumi 'leaving'
 - ▶ ϵ : non-zero energy generation rate
 - ▶ For the Sun, average rate $\epsilon_{\odot}$

- ▶ Gives,

$$\frac{dL}{dM} = \epsilon$$

- ▶ In terms of r ,

$$\frac{dL}{dr} = 4\pi\rho r^2\epsilon$$



Third Equation of Stellar Structure

LIBERATION OF GRAVITATIONAL ENERGY

- ▶ Gravitational potential energy, $U = -G \frac{Mm}{r}$

- ▶ As $r \rightarrow 0$, U becomes more negative
 - ▶ Suggests E converted to other forms, eg. K.E.

- ▶ Total work done: $W = - \int_0^M \frac{GMdm}{r}$

- ▶ If ρ uniform, $M \propto r^3$

$$W = - \frac{3}{5} \frac{GM^2}{R} = - \alpha \frac{GM^2}{R}$$

- ▶ Where $\alpha \sim 1$
- ▶ If energy output were constant

$$t_{KH} \equiv - \frac{W}{L} = \frac{\alpha GM}{\epsilon R}$$

Kelvin-Helmholtz timescale (~P2.59)

- ▶ For the Sun, $\sim 10^{15} \text{s} \sim 30 \text{ Myr}$
- ▶ But we find fossils 100s of millions years old!

NUCLEAR TIMESCALE

- Conversion of mass to energy in fusion processes

$$\Delta E = \varepsilon M c^2$$

- ε is rough typical fraction of rest mass lost to energy in nuclear reactions
- This will be product of mass conversion fraction in fusion: **0.7%** for Hydrogen fusion
 - And factor for what fraction of star's mass ultimately fused: **~10%**
- For the Sun, energy provided from nuclear reactions:
 - $\Delta E_{nuc} = 0.1 \times 0.007 \times M_{\odot} c^2 = 1.3 \times 10^{44} \text{J}$
- Rate of conversion of mass is L_{nuc}
 - TE gives $L = L_{nuc}$
 - $t_{nuc} \approx \frac{\varepsilon M c^2}{L} \sim 10^{10} \text{yr}$
 - More generally, and in solar units,
- Sets timescale of stellar ages and evolution

$$t_{nuc} \approx \varepsilon 4.5 \times 10^{20} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right) \quad (\text{P2.62})$$

STARS TRANSPORT ENERGY

- ▶ 3 equations of stellar structure: Four unknowns: $M(r)$, $\rho(r)$, $L(r)$, $T(r)$
- ▶ Consider the source of energy
 - ▶ Nuclear reactions in core
 - ▶ Produced photons, $e^\pm$ heat the core
 - ▶ For Sun, neutrinos carry away 2.3% of energy
- ▶ Energy generated: how get to surface?
 - ▶ **Conduction**: Energy transport thru particle collisions
 - ▶ **Radiation**: Energy transport thru radiation, absorption, remission of photons
 - ▶ **Convection**: Energy transport thru motion of heated material
- ▶ Generalized Energy flux (energy/time/area):
 - ▶ D : diffusivity
 - ▶ U : energy density

$$\vec{F} = -D \vec{\nabla} U$$

CONDUCTIVE ENERGY TRANSPORT

- ▶ Heat flow from particle random thermal motion

- ▶ To conduct from core to surface,

$$t_{cond} \sim \frac{R_{\odot}^2}{D_{th}}$$

- ▶ Treat star as a dense gas of pure, ionized H

- ▶ Thermal conductivity primarily fast electrons,

$$D_{th} = \frac{1}{3} v_{th} \lambda_e$$

- ▶ v_{th} : random thermal speed of free electrons

- ▶ λ_e : electron mean free path

- ▶ Thermal velocity

- ▶ At temperature, T, can express

$$\frac{v_{th}}{c} = \left(\frac{3kT}{m_e c^2} \right)^{1/2}$$

- ▶ Central temperature $\sim 10^7$ K

- ▶ Pick temperature halfway between this and surface temperature

$$\frac{v_{th}}{c} = 0.05 \sqrt{\frac{T}{5 \times 10^6 \text{ K}}}$$

ELECTRON MEAN FREE PATH

- ▶ Take a single atom with radius $2a_0$ and velocity, v
 - ▶ Move a distance vt in time t
 - ▶ Sweep up volume $V = \pi(2a_0)^2 vt = \sigma vt$
 - ▶ Term σ ($= \pi(2a_0)^2$) the (classical) 'cross section'
 - ▶ In quantum mechanics
 - ▶ replaced by a probability w/units of area

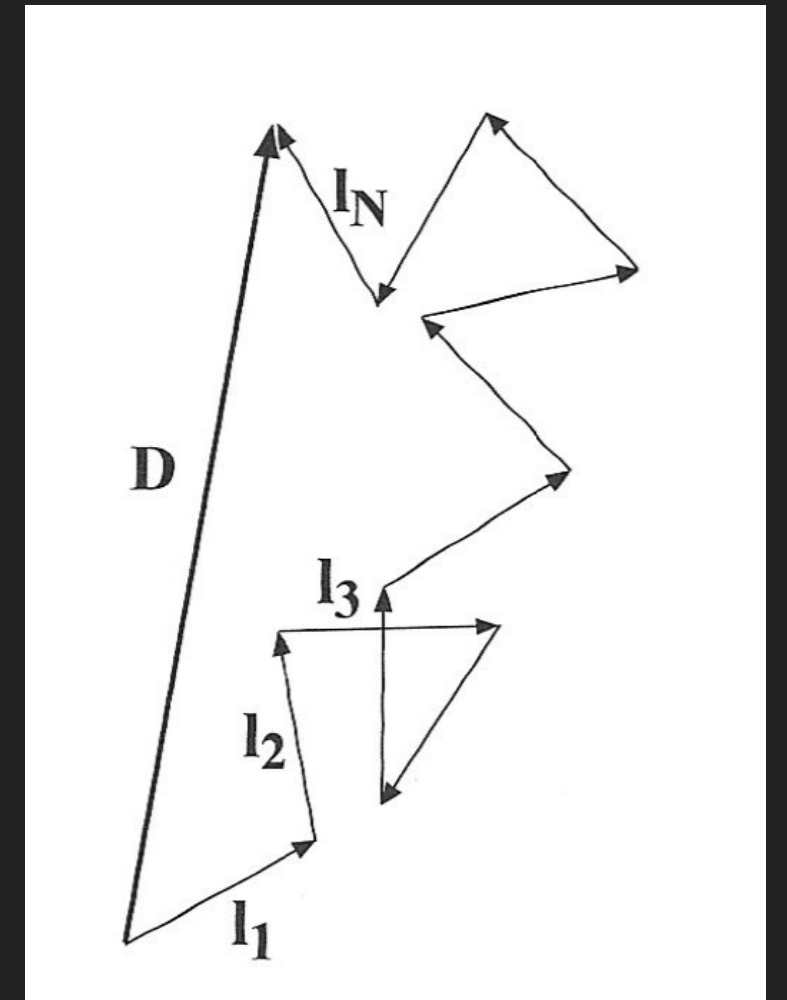
- ▶ Average distance traveled between collisions

$$\lambda_e = \frac{vt}{\text{\#atoms collided}} = \frac{vt}{n\sigma vt} = \frac{1}{n\sigma} \quad \text{Mean free path}$$

- ▶ n : # density of objects free electron can collide with

$$n = n_e + n_p = \frac{2\rho}{m_p} = 1.69 \times 10^{24} \text{ cm}^{-3} \left(\frac{\rho}{\rho_\odot} \right)$$

- ▶ σ : cross section for collisions



ESTIMATING CROSS SECTION FOR COLLISIONS

- ▶ When electron pushed to 90 deg., $\Delta E \sim$ initial KE

$$\frac{\rho^2}{r_e} \approx \frac{1}{2} m_e v_{th}^2$$

- ▶ Then effective cross section

$$\sigma \approx \pi r_e^2 \approx 4\pi \left(\frac{e^2}{m_e v_{th}^2} \right)^2$$

- ▶ We factor this as

$$\begin{aligned} \sigma &\approx \frac{3}{2} \sigma_e \left(\frac{v_{th}}{c} \right)^{-4} \\ &\approx 1.6 \times 10^{-19} \text{ cm}^2 \left(\frac{v_{th}/c}{0.05} \right)^{-4} \end{aligned}$$

- ▶ Where we have

$$\begin{aligned} \sigma_e &= \frac{8\pi}{3} \left(\frac{e^2}{m_e v_{th}^2} \right)^2 \\ &= 6.652 \times 10^{-25} \text{ cm}^2 = 0.6652 \text{ barn} \end{aligned}$$

Thomson
cross section

CALCULATING CONDUCTION TIME

$$\lambda_e = \frac{m_p}{2\rho\sigma} \approx 3.7 \times 10^{-6} \text{ cm} \left(\frac{\rho}{\rho_\odot} \right)^{-1} \times \left(\frac{v_{th}/c}{0.05} \right)^4$$

- ▶ We can now calculate thermal diffusivity

$$D_{th} = \frac{1}{3} v_{th} \lambda_e$$
$$\approx 1900 \text{ cm}^2 \text{ s}^{-1} \left(\frac{\rho}{\rho_\odot} \right)^{-1} \left(\frac{v_{th}/c}{0.05} \right)^5$$

- ▶ If we take the average $v_{th}/c \sim 0.05$
- ▶ Yields $D_{th} \approx 50 \text{ cm}^2 \text{ s}^{-1}$
- ▶ Conduction time is then

$$t_{cond} \sim \frac{R_\odot^2}{D_{th}} \sim 2 \times 10^{19} \text{ s} \sim 4 \times 10^2 \text{ Gyr}$$

- ▶ This is 40x expected lifetime of sun, and 80x known age of Solar System

RADIATIVE ENERGY TRANSPORT

- ▶ Problem with conduction
 - ▶ Propagated by electrons
 - ▶ Move slowly
 - ▶ Short mean free path
 - ▶ Low diffusivity
- ▶ Consider interior filled with photons
 - ▶ Interior not transparent,
$$\lambda_\gamma = \frac{1}{n\sigma}$$
 - ▶ n : # density of absorbers and scatterers
 - ▶ σ : typical cross section for interactions with photons
 - ▶ Can also be written as,
$$\lambda_\gamma = \frac{1}{\rho\kappa}$$
 - ▶ κ : opacity of stellar material
 - ▶ Units are area/unit mass
 - ▶ Sum of cross sections of all absorbers and scatterers in volume
 - ▶ Divided by mass in volume

RADIATION DIFFUSIVITY

- ▶ Diffusion equation,
 - ▶ At radius, r , photon black body energy density,
 - ▶ Where $a = 7.566 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$

$$\vec{F} = -D_\gamma \vec{\nabla} U_\gamma$$

$$U_\gamma(r) = aT(r)^4$$

- ▶ We can write diffusivity as,

$$D_\gamma = \frac{1}{3} c \lambda_\gamma = \frac{3}{3\kappa\rho}$$

- ▶ Note that $c \gg v_{th}(e)$ so this helps!
- ▶ Mean free path for photons,
 - ▶ Other forms of scattering than just Thomson
 - ▶ Lower bound on σ means upper bound on λ_γ

$$\lambda_\gamma = \frac{1}{n_e \sigma_e} = \frac{m_p}{\rho \sigma_e} = 1.8 \text{ cm} \left(\frac{\rho}{\rho_\odot} \right)^{-1}$$

- ▶ Diffusivity then

$$D_\gamma = 2 \times 10^{10} \text{ cm}^2 \text{ s}^{-1} \left(\frac{\lambda_\gamma}{2 \text{ cm}} \right)$$

- ▶ Timescale for diffusion

$$t_\gamma \sim \frac{R_\odot^2}{D_\gamma} \sim 8000 \text{ yr} \left(\frac{\lambda_\gamma}{2 \text{ cm}} \right)^{-1}$$

- ▶ This is short on the timescale of a star's life!

RADIATIVE DIFFUSION EQUATION

- ▶ In general

$$\frac{L(r)}{4\pi r^2} = -D_\gamma \frac{dU_\gamma}{dr} = -D_\gamma \frac{d}{dr}(aT^4)$$

- ▶ Substitute in for D_γ

$$\frac{L(r)}{4\pi r^2} = -\frac{c}{3\kappa\rho}(a)4T^3 \frac{dT}{dr}$$

- ▶ So that

$$\frac{dT}{dr} = -\frac{3\kappa\rho L(r)}{16\pi acT^3 r^2}$$

4th Equation of
Stellar Structure

- ▶ Now we have four equations in four unknowns!

- ▶ $M(r), L(r), T(r), \rho(r)$

- ▶ We also have four constitutive relations

- ▶ $\kappa(\rho, T, X_i)$

- ▶ $\epsilon(\rho, T, X_i)$

- ▶ $P(\rho, T, X_i)$

- ▶ Need to identify functional forms for these

SUPPLEMENTAL MATERIAL

ROTATIONAL AND MAGNETIC FORCES

- ▶ Consider Sun

- ▶ Rotation: 27 d

- ▶ KE of rotation relative to gravitational binding Energy

$$\frac{KE_{rot}}{PE_{grav}} \sim \frac{M\omega^2 R^2}{GM^2/R} = \frac{\omega^2 R^3}{GM} \left(\sim 2 \times 10^{-5} \text{ for Sun} \right)$$

- ▶ Magnetic field mT to few x 0.1 T

- ▶ Ratio of B field energy density to gravitational

- ▶ assume $B = 0.1 \text{ T}$, which is the value in sunspots

$$\frac{B^2 / 2\mu_o}{GM^2 / R^4} = \frac{B^2 R^4}{2\mu_o GM^2} \left(\sim 10^{-11} \text{ for Sun} \right)$$

- ▶ So assumptions pretty good for Sun-like stars

DYNAMICAL TIMESCALE – PRIALNIK

- ▶ Change in structure of star
 - ▶ Spherical symmetry
 - ▶ Characteristic radius, R
 - ▶ Dynamic rate of change due to gravity
 - ▶ Expansion and contraction
 - ▶ Too much or not enough pressure balancing gravity

- ▶ Consider escape velocity, $v_{esc} = \sqrt{2GM/R}$

$$\text{▶ } t_{dyn} \approx \frac{R}{v_{esc}} = \sqrt{\frac{R^3}{2GM}} = \sqrt{\frac{1}{2G} \frac{3}{4\pi} \left(\frac{4\pi R^3}{3M} \right)} \approx O\left(\frac{1}{\sqrt{G\rho}}\right) \quad (\text{P2.57})$$

$$\text{▶ For the Sun, } \sim 1000\text{s}, \quad t_{dyn} \simeq 1000 \sqrt{\left(\frac{R}{R_\odot}\right)^3 \left(\frac{M_\odot}{M}\right)} \text{ sec.} \quad (\text{P2.58})$$

- ▶ Many orders $\ll$ than stellar ages (4.6 billion years for Sun)

FORCES ACTING

- ▶ From Newton's 2nd law
 - ▶ Momentum conservation law for body of fixed mass

$$F = ma \rightarrow dF = dm \frac{d^2 r}{dt^2} = F_g + F_{p,t} + F_{p,b}$$

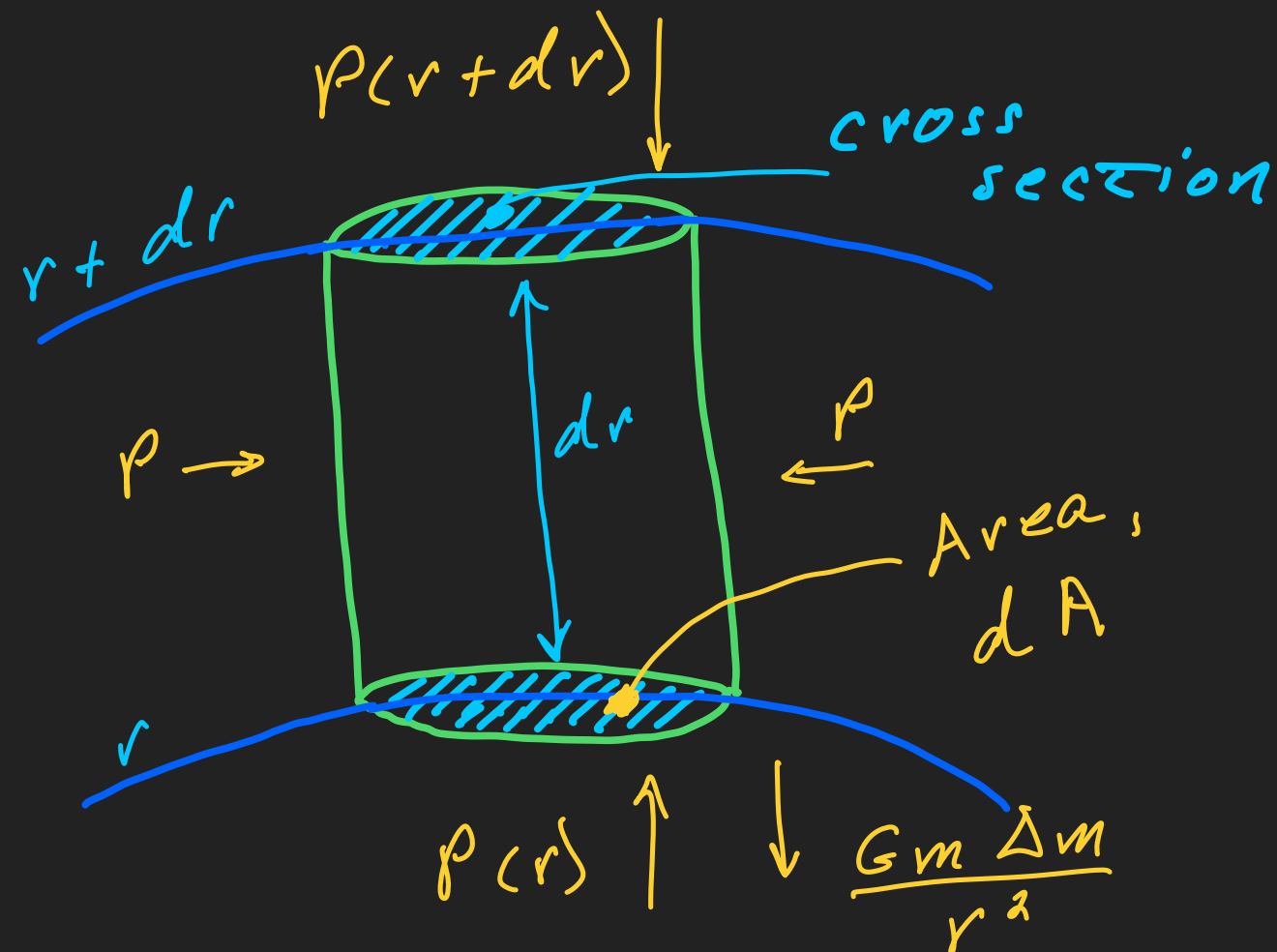
- ▶ Change in pressure over dx

$$F_{p,t} = -(F_{p,b} + dF_p)$$

This means $dm \frac{d^2 r}{dt^2} = F_g - dF_p$ where

$$F_g = -G \frac{M_r dm}{r^2}$$

- ▶ Where M_r is mass inside sphere of radius, r



THE EQUATION OF MOTION

- ▶ The value of dF_p is just area times the dP over the cylinder, giving

$$dm \frac{dr^2}{dt^2} = -G \frac{M_r dm}{r^2} - AdP$$

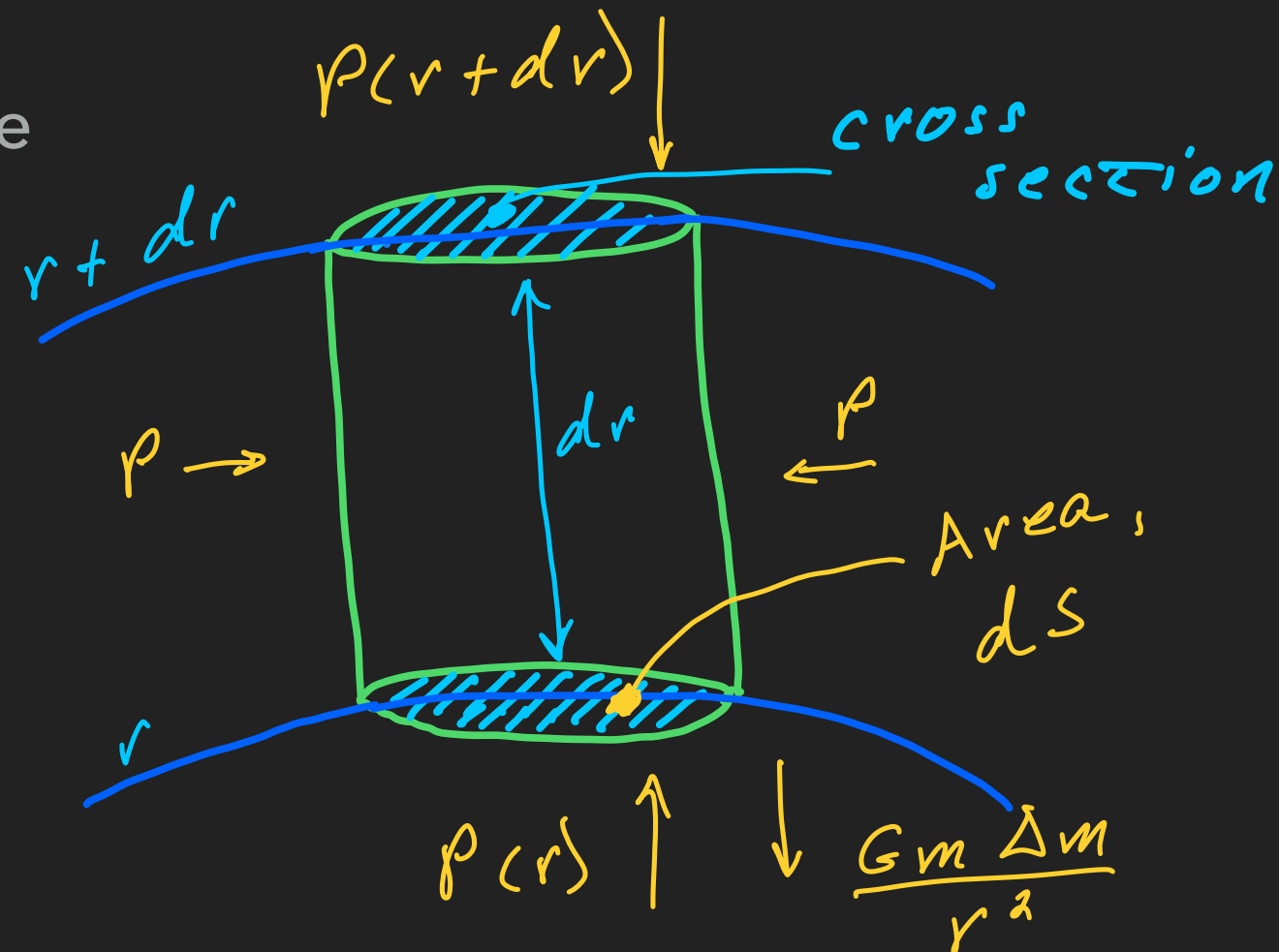
- ▶ For constant density, $dm = \rho A dr$

$$\rho A dr \frac{dr^2}{dt^2} = -G \frac{M_r \rho A dr}{r^2} - AdP$$

- ▶ Dividing by $\rho A dr$, we obtain

$$\ddot{r} = \frac{-GM_r}{r^2} - \frac{1}{\rho} \frac{\partial P}{\partial r} \quad (\text{P2.12})$$

- ▶ Equation of Motion

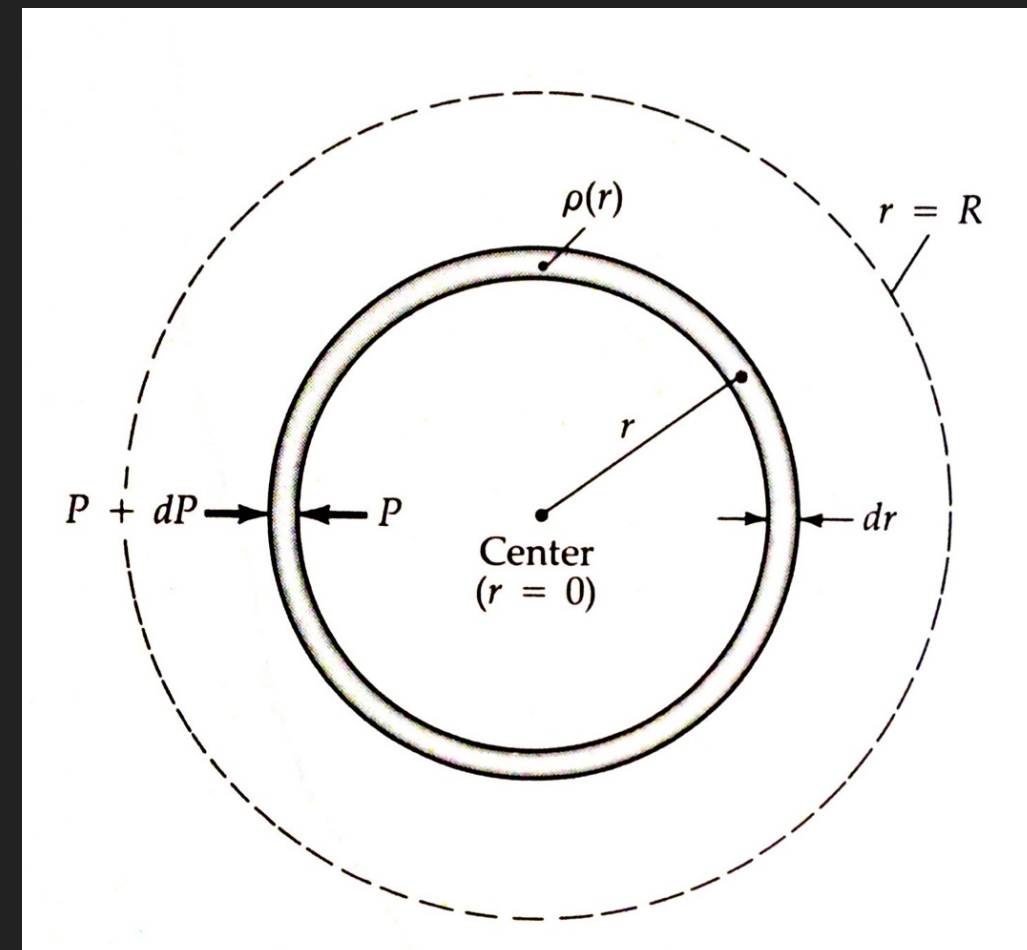


CONDITION FOR HYDROSTATIC EQUILIBRIUM

- ▶ When acceleration negligible
 - ▶ I.e. equilibrium condition
 - ▶ First term is zero.
 - ▶ we obtain

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2}$$

Condition for Hydrostatic Equilibrium
(P2.14)



PRESSURE AT CENTER OF STAR

- ▶ Integrate from center to surface

$$P(M) - P(0) = - \int_0^M \frac{Gm \, dm}{4\pi r^4}$$
$$P_c = \int_0^M \frac{Gm \, dm}{4\pi r^4} > \int_0^M \frac{GM \, dm}{4\pi R^4}$$

- ▶ Where R is radius of the star

$$P_c > \frac{GM^2}{8\pi R^4} = 4.4 \times 10^{13} \left(\frac{M}{M_\odot} \right)^2 \left(\frac{R_\odot}{R} \right)^4 \text{Nm}^{-2} \quad (\text{P2.18})$$

- ▶ Sun: 4.5×10^8 atmospheres!

THE VIRIAL THEOREM

- ▶ HE establishes link between gravitational potential energy and internal energy

- ▶ Or KE in system of free particles

- ▶ Take Eq. Of HE x volume ($= 4\pi r^3/3$) and integrate over whole star

$$\int_0^{P(R)} V dP = -\frac{1}{3} \int_0^M \frac{Gm dm}{r} \left(= \frac{1}{3} \Omega \right)$$

grav. p E of star

- ▶ Integrate left side by parts,

$$\int_0^{P(R)} V dP = [PV]_0^R - \int_0^{V(R)} P dV$$

- ▶ Note that $P=0$ at surface, and $V=0$ at center, so first term yields zero

$$-\int_0^{V(R)} P dV = -\frac{1}{3} \int_0^M \frac{Gm dm}{r} = \frac{1}{3} \Omega$$

- ▶ Using $dV = dm/\rho$

$$\Omega = -3 \int_0^M \frac{P}{\rho} dm \quad (\text{P2.23})$$

General Virial
Theorem

GRAVITATION AND KELVIN-HELMHOLTZ TIMESCALE

- ▶ Gravitational force on point mass dm

- ▶ Outside spherically symmetric mass M_r , $dF_{g,i} = G \frac{M_r dm_i}{r^2}$

- ▶ Implies $dU_{g,i} = -G \frac{M_r dm_i}{r}$

- ▶ Consider a shell of thickness dr , $dm = 4\pi r^2 \rho dr$

$$dU_g = -G \frac{M_r 4\pi r^2 \rho}{r} dr$$

- ▶ Integrating from center to surface

$$U_g = -4\pi G \int_0^R M_r \rho r dr. \quad (\text{R is radius of star})$$

GRAVITATIONAL POTENTIAL ENERGY

- ▶ To do this exactly, we'd need the density profile
 - ▶ Assume constant density

$$\rho \sim \bar{\rho} = \frac{M}{\frac{4}{3}\pi R^3} \quad \rightarrow \quad M_r \sim \frac{4}{3}\pi r^3 \bar{\rho}$$

- ▶ This gives us

$$\begin{aligned} \blacktriangleright \quad U_g &\sim -4\pi G \int_0^R \frac{4}{3}\pi r^3 \bar{\rho} \bar{\rho} r dr \\ \blacktriangleright \quad &= -\frac{(4\pi)^2}{3} G \bar{\rho}^2 \int_0^R r^4 dr \\ \blacktriangleright \quad &= -\frac{16}{15} \pi^2 G \bar{\rho}^2 R^5 = -\frac{16}{15} \pi^2 G \left(\frac{M}{\frac{4}{3}\pi R^3} \right)^2 R^5 \\ \blacktriangleright \quad &= -\frac{3}{5} \frac{GM^2}{R} \end{aligned}$$

KELVIN-HELMHOLTZ TIMESCALE

- ▶ Virial theorem means half goes to KEs: $E \sim \frac{3}{10} \frac{GM^2}{R}$
- ▶ Consider infalling Sun from $R \gg R_\odot$
 - ▶ Total gravitational energy liberated

$$\Delta E_g = - (E_f - E_i) \sim E_f \sim \frac{3}{10} \frac{GM_\odot^2}{R_\odot}$$
 - ▶ This is about $1.1 \times 10^{41} \text{J}$ for the Sun
- ▶ To output at that rate approximately constantly

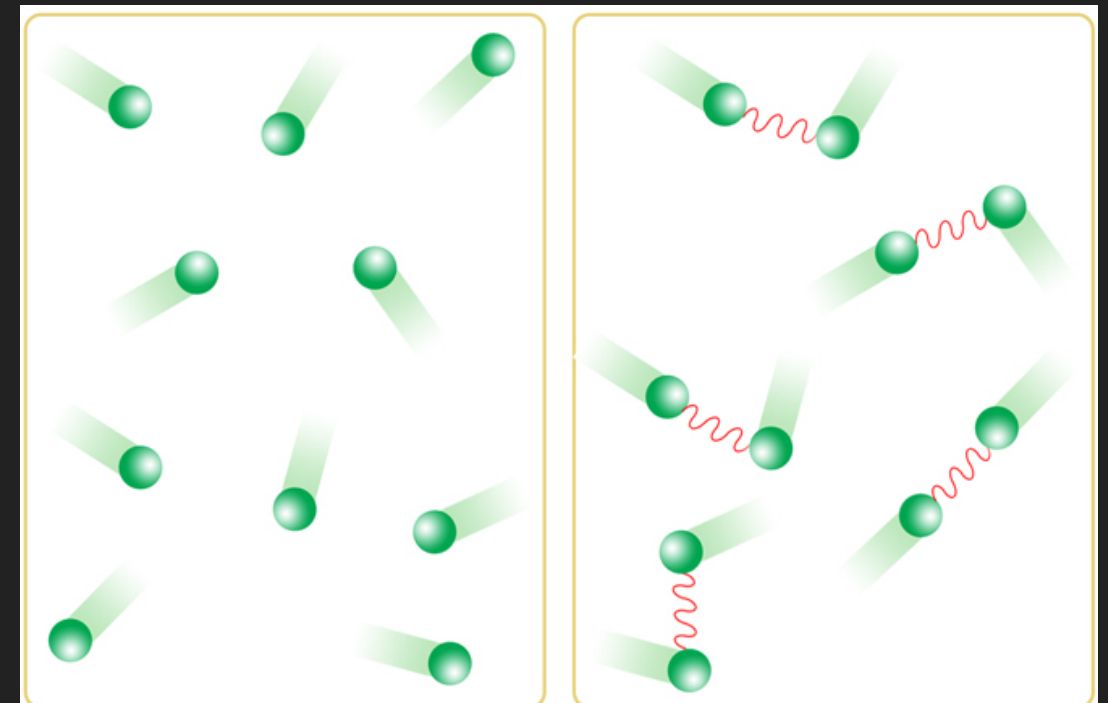
$$t_{KH} = \frac{\Delta E_g}{L_\odot} \sim 10^7 \text{yrs} \quad \text{Kelvin-Helmholtz timescale} \quad (\sim \text{P2.59})$$
 - ▶ $\ll$ age of Sun (4.6 by)
 - ▶ Still way too short for evolution
- ▶ Relative to Sun:

$$\tau_{th} \approx 10^{15} \left(\frac{M}{M_\odot} \right)^2 \left(\frac{R_\odot}{R} \right) \left(\frac{L_\odot}{L} \right) \text{s} \quad (\text{P2.60})$$

IDEAL GAS

- ▶ Many kinds of temperature in a gas
 - a) Effective T: from Stefan-Boltzmann, specific level of star
 - b) Excitation T: defined by Boltzmann Eq.
 - c) Ionization T: defined by Saha Eq.
 - d) Kinetic T: contained in Maxwell-Boltzmann velocity distribution
 - e) Color T: from fit to star continuum to Planck blackbody function
- ▶ Consider a gas
 - ▶ Randomly moving point-like particles
 - ▶ no inter particle interactions
 - ▶ Just elastic collisions
 - ▶ Gas particles and blackbody radiation in equilibrium
- ▶ Such a gas is an 'ideal gas'
 - ▶ Temperatures b) thru e) all same in this case

Ideal Gas vs. Real Gas



IDEAL GAS EQUATION OF STATE

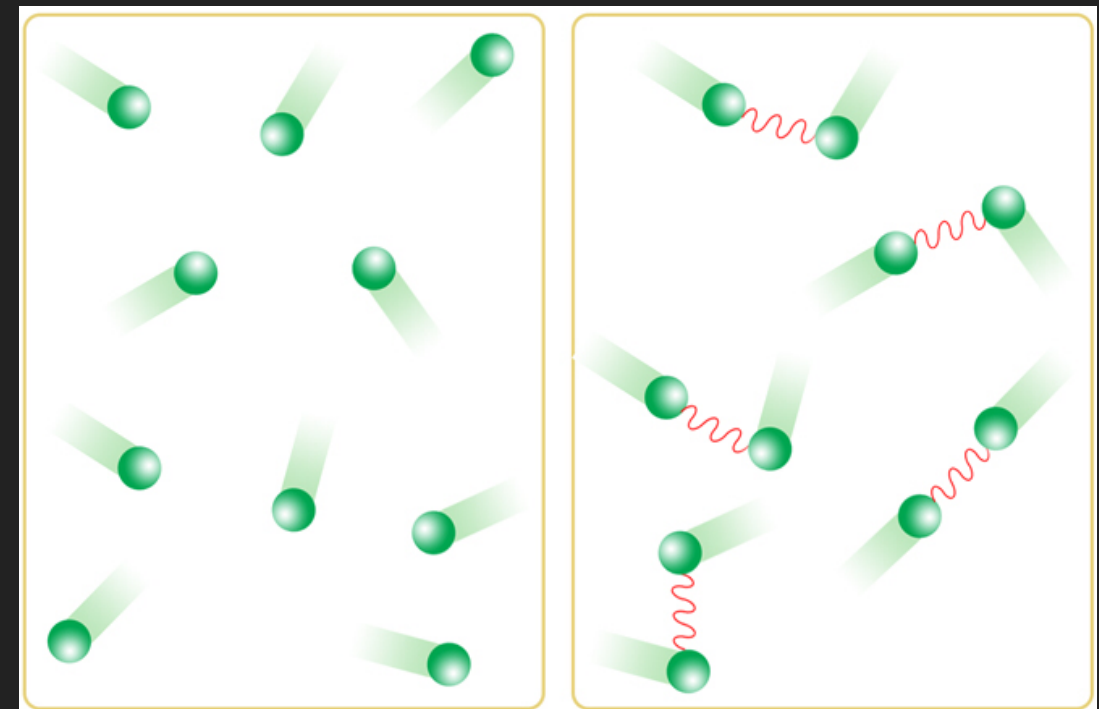
- ▶ associated pressure:

$$PV = NkT$$

ideal gas law

- ▶ V is volume of gas
 - ▶ N is # of particles in volume
-
- ▶ If temperature decreases: lower pressure → contraction
 - ▶ Increasing temperature: higher pressure → expansion

Ideal Gas vs. Real Gas



$$\Omega = -3 \int_0^M \frac{P}{\rho} dm$$

IDEAL GAS

► Density ρ , temperature T

► Gas particle mass m_g

► Gas pressure. $P = \left(\frac{\rho}{m_g} \right) kT$

► KE per particle. $= \frac{3}{2} kT$

► So internal energy per unit mass $u = \frac{3}{2} \frac{kT}{m_g} = \frac{3}{2} \frac{P}{\rho}$

► Using this and Virial theorem, obtain

► total internal energy, $U = \int_0^M u dm = -\frac{1}{2} \Omega$

$$U = \int_0^M \frac{3}{2} \frac{kT}{m_g} dm = \frac{3}{2} \frac{k\bar{T}}{m_g} M \quad (\text{P2.28})$$

Bozma's const.

EXTRACTING TEMPERATURE

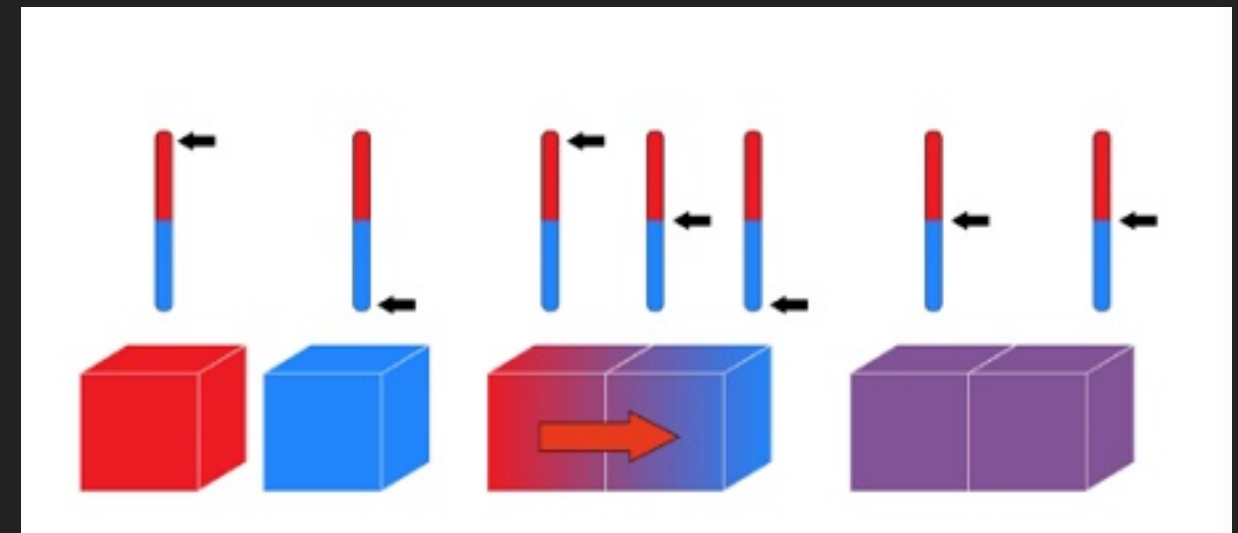
- ▶ Gravitational PE $\Omega = -\alpha \frac{GM^2}{R}$
- ▶ Equating and isolating T, we obtain: $\frac{1}{2} \left(\alpha \frac{GM^2}{R} \right) = \frac{3}{2} \frac{k\bar{T}M}{m_g}$
- $\Rightarrow \bar{T} = \frac{\alpha}{3} \frac{Gm_g}{k} \frac{M}{R}$ (P2.29)
- ▶ Consider average density $\bar{\rho} = \frac{M}{V} = \frac{M}{4\pi R^3/3}$ $\bar{T} \propto M^{2/3} \bar{\rho}^{1/3}$
 - ▶ Star of the same mass – is hotter if denser
- ▶ Further consider gas of hydrogen, $\bar{T} \approx 4 \times 10^6 \left(\frac{M}{M_\odot} \right) \left(\frac{R_\odot}{R} \right) \text{ K}$
 - ▶ $\gg T_{\text{eff}}$
 - ▶ Further inside must be hotter still
 - ▶ H, He and many metals ionized
 - ▶ Material of a star is a plasma

DISCUSSION QUESTION

Assuming a ball of pure hydrogen, how does the average temperature of a star vary with stellar radius?

THERMAL EQUILIBRIUM

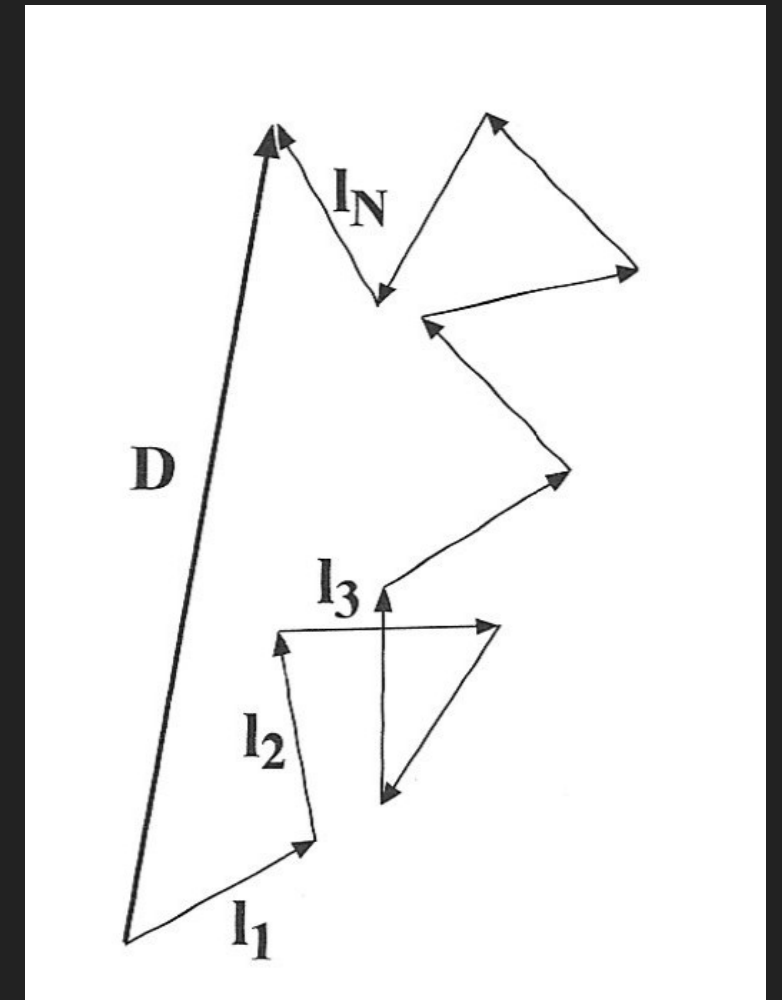
- ▶ For ideal gas
 - ▶ No net. flow of energy thru box
 - ▶ Or between matter and radiation
 - ▶ **Thermodynamic equilibrium**
 - ▶ Same amount of energy coming into a region (eg. Photosphere) and radiating away
- ▶ Consider a star
 - ▶ Emit radiation, flow of energy outward
 - ▶ Do not have global TE
 - ▶ In a box:
 - ▶ Particles and. Photons arrive from
 - ▶ Hotter or cooler regions



MATTER AND RADIATION

- ▶ A photon 'gas'
 - ▶ Each particle carries quantum of E: $h\nu$
 - ▶ Momentum $h\nu/c$
 - ▶ Star made up of mixture of
 - ▶ ions, electrons, photons
- ▶ Components not always at same temperature
 - ▶ Solar wind does not satisfy
 - ▶ Protons $T \sim 10^6 K$
 - ▶ Electrons. $2 \times T$ of protons
 - ▶ Radiation 6000 K
- ▶ Frequent collisions
 - ▶ Between photons and matter particles
 - ▶ Leads toward equilibrium

- ▶ Collisions between matter particles and photons
- ▶ Radiation becomes blackbody when equilibrium between matter and radiation
 - ▶ Photon E: Planck function
 - ▶ Temperature of gas and radiation the same
 - ▶ Single T valid over a distance much larger than the typical travel distance ('mean free path') of particles
- ▶ Nearly blackbody in stellar interiors
 - ▶ So gas T = radiation T at each point
 - ▶ Local unique temperature



PURE HYDROGEN GAS

- ▶ Assume ideal gas

- ▶ # hydrogen atoms per cubic meter: $n = \frac{\rho}{m_H} = 1.25 \times 10^{23} \text{ m}^{-3}$

- ▶ Take a single atom with radius $2a_0$ and velocity, v

- ▶ Move a distance vt in time t

- ▶ Sweep up volume $V = \pi(2a_0)^2 vt = \sigma vt$

- ▶ Term σ ($= \pi(2a_0)^2$) the (classical) 'cross section'

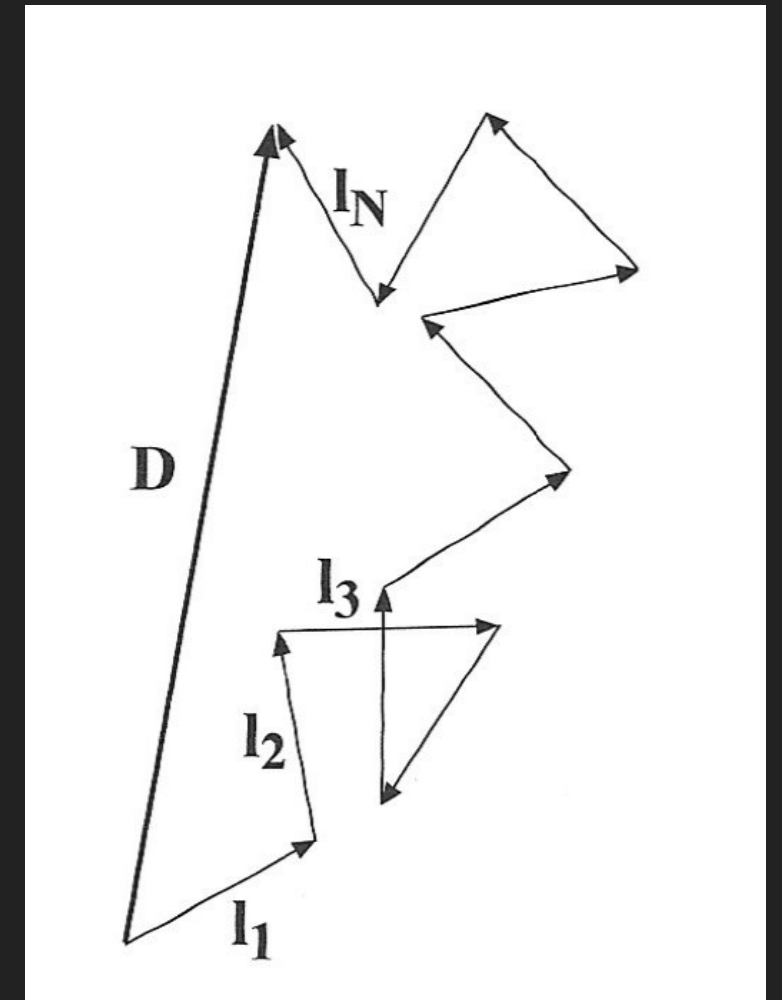
- ▶ In quantum mechanics

- ▶ replaced by a probability
 - ▶ units of area

- ▶ Average distance traveled between collisions

$$\ell = \frac{vt}{\text{\#atoms collided}} = \frac{vt}{n\sigma vt} = \frac{1}{n\sigma}$$

Mean free path



MEAN FREE TIME

- ▶ Time between collisions
- ▶ When $\ll$ timescale for change of macroscopic properties
 - ▶ TE holds
 - ▶ Note: temperatures may change with time
 - ▶ This is scenario in stars

LOCAL THERMODYNAMIC EQUILIBRIUM (LTE)

- ▶ If $\ell \ll$ temperature scale height
 - ▶ Local Thermodynamic Equilibrium
 - ▶ Particles and photons
 - ▶ Cannot escape local environment
 - ▶ Confined to limited volume of nearly constant temperature
- ▶ Allows calculation of all thermodynamic properties
 - ▶ T, density, composition
 - ▶ At a radius within star
 - ▶ Stellar structure for star of given mass
 - ▶ Composition can be given by mass fraction of species within star

- Uniquely determined @ time t by T, density, composition at each point within

ENERGY EQUATION

- ▶ Internal energy of a system changed by
 - ▶ Heat
 - ▶ Work → involves change in volume
- ▶ Consider small element in star
 - ▶ T, density, composition constant
 - ▶ Spherical shell from r to $r+dr$
 - ▶ Volume $dV = 4\pi r^2 dr$
 - ▶ Mass $dm = \rho dV = \rho 4\pi r^2 dr$ (P2.1)

WHAT IS WORK DONE?

- ▶ Let 'u' be internal energy per unit mass
- ▶ P is pressure
- ▶ δf Is change in any quantity within mass element over time, δt

$$\delta(u \, dm) = dm \, \delta u = \delta Q + \delta W$$

- ▶ $\delta Q \rightarrow$ heat absorbed or emitted
- ▶ $\delta W \rightarrow$ work done on dm [dm is constant – conservation of mass]

$$\delta W = -P\delta V = -P\delta\left(\frac{dV}{dm}dm\right) = -P\delta\left(\frac{1}{\rho}\right)dm$$

- ▶ Compression $\rightarrow$ shrink element volume, $dV < 0$
 - ▶ Addition of energy
- ▶ Expansion $\rightarrow dV > 0$ at expense of element's energy

SOURCES OF HEAT

- ▶ Release of nuclear energy

- ▶ Rate of release per unit mass, 'q'

- ▶ Balance of heat into and out of volume element

- ▶ Heat flowing $\perp$ to spherical surface, $F(m)$. $F(M) \equiv L$

$$\delta Q = q dm \delta t + F(m) \delta t - F(m + dm) \delta t$$

- ▶ However $F(m + dm) = F(m) + \left(\frac{\partial F}{\partial m} \right) dm$

$$\therefore \delta Q = \left(q - \frac{\partial F}{\partial m} \right) dm \delta t$$

- ▶ By substitution

$$dm \delta u + P \delta \left(\frac{1}{\rho} \right) dm = \left(q - \frac{\partial F}{\partial m} \right) dm \delta t$$

- ▶ Divide by $dm \delta t$ and take $\delta t \rightarrow 0$

$$\dot{u} + P \frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) = q - \frac{\partial F}{\partial m} \quad (\text{P2.6})$$

THERMAL EQUILIBRIUM

- ▶ Temporal derivatives vanish

$$q = dF/dm$$

- ▶ Integrate over mass

$$\int_0^M q \, dm = \int_0^M dF = L$$

- ▶ First term is nuclear luminosity, L_{nuc}

- ▶ Implies $L = L_{nuc}$

- ▶ Energy radiated away at same rate as produced via nuclear reactions

TOTAL ENERGY OF A STAR

$$\triangleright dV = 4\pi r^2 dr$$

$$\triangleright dm = \rho dV = \rho 4\pi r^2 dr \quad (\text{P1.5})$$

$$\triangleright \dot{u} + P \frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) = q - \frac{\partial F}{\partial m} \quad (\text{P2.6})$$

► Integrate energy equation (2.6) over whole star,
$$\int_0^M \dot{u} dm + \int_0^M P \frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) dm = L_{nuc} - L$$

►
$$\frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) = \frac{\partial}{\partial t} \left(\frac{\partial V}{\partial m} \right) = \frac{\partial \dot{V}}{\partial m} = \frac{\partial(4\pi r^2 \dot{r})}{\partial m}$$

► Which leads for the 2nd term to
$$\int_0^M P \frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) dm = \int_0^M P \frac{\partial(4\pi r^2 \dot{r})}{\partial m} dm = \int_0^M P \frac{\partial \dot{V}}{\partial m} dm$$

► For the first term, swap independent operators,
$$\int_0^M \dot{u} dm = \frac{d}{dt} \int_0^M u dm = \dot{U}$$

► Integrating 2nd term by parts

$$\int_0^M P \frac{\partial \dot{V}}{\partial m} dm = \left[P \dot{V} \right]_0^M - \int_0^M 4\pi r^2 \dot{r} \frac{\partial P}{\partial m} dm$$

► Recall $P=0$ at the surface, and $\dot{V}=0$ at the center: so the first term is zero

► We now have
$$\dot{U} - \int_0^M 4\pi r^2 \dot{r} \frac{\partial P}{\partial m} dm = L_{nuc} - L \quad (\text{P2.36})$$

$$\dot{r} = -\frac{Gm}{r^2} - 4\pi r^2 \frac{\partial P}{\partial m} \quad (\text{P2.13})$$

RATE OF CHANGE OF TOTAL STELLAR ENERGY

- ▶ Multiply equation of motion (2.13) by $\dot{r}$ and integrate over whole star

$$\int_0^M \dot{r} \ddot{r} dm = - \int_0^M \frac{Gm}{r^2} \dot{r} dm - \int_0^M 4\pi r^2 \dot{r} \frac{\partial P}{\partial m} dm$$

- ▶ Rearranging derivatives on left, $\int_0^M \dot{r} \ddot{r} dm = \int_0^M \frac{d}{dt} \left(\frac{1}{2} \dot{r}^2 \right) dm = \frac{d}{dt} \int_0^M \frac{1}{2} \dot{r}^2 dm = \dot{K}E$

- ▶ On right hand, 1st term, $-\int_0^M Gm \frac{\dot{r}}{r^2} dm = \int_0^M Gm \left(\frac{1}{r} \right) \dot{} dm = \frac{d}{dt} \int_0^M \frac{Gm dm}{r} = -\dot{\Omega}$

- ▶ So $\dot{K} + \dot{\Omega} = - \int_0^M 4\pi r^2 \dot{r} \frac{\partial P}{\partial m} dm \quad (\text{P2.41})$

$$\therefore \dot{U} + \dot{K} + \dot{\Omega} (= \dot{E}) = L_{nuc} - L \quad (\text{P2.42})$$

- ▶ This is **rate of change of total stellar energy**

$$\triangleright U = \frac{3}{2} \frac{k}{m_g} \bar{T} M \quad (\text{P2.28})$$

TRENDS IN EQUILIBRIUM

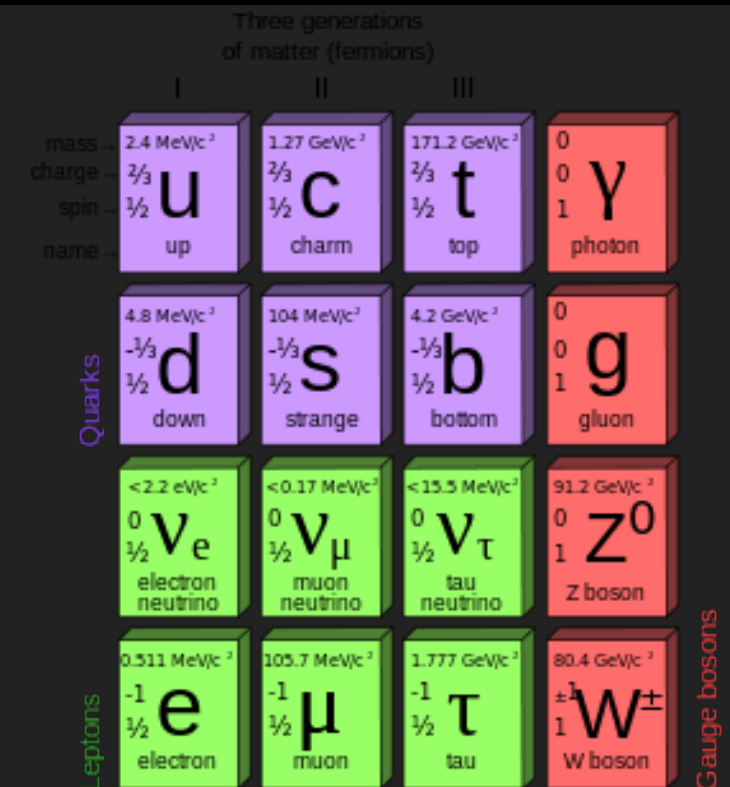
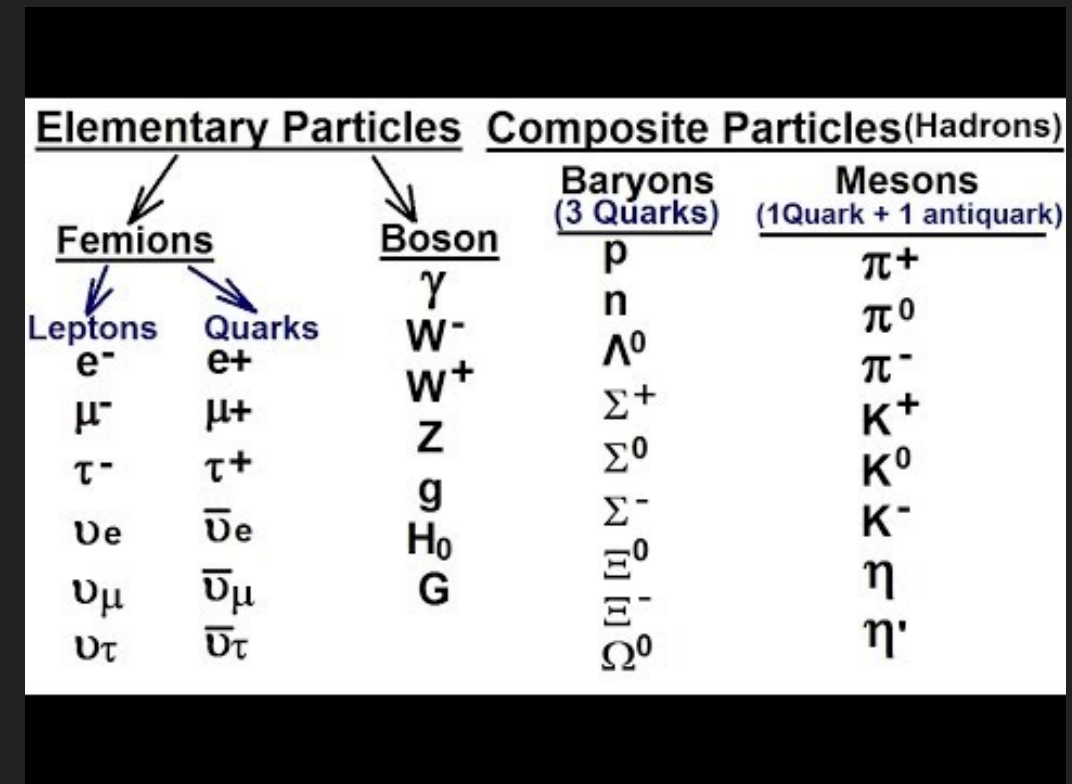
- ▶ Thermal equilibrium, $\dot{E} = 0$
- ▶ Hydrostatic equilibrium, $K \rightarrow 0$
- ▶ U and Ω related by virial theorem
 - ▶ Either determines total E of star
 - ▶ So U and Ω Conserved, not just sum
- ▶ Star cannot cool throughout and expand
- ▶ Star has negative heat capacity
 - ▶ Gets hotter when losing energy
 - ▶ From virial theorem, $E = U + \Omega = \frac{1}{2}\Omega = -U$
 - ▶ If $\dot{E} < 0$, then U must increase (so must T , via P2.28)
 - ▶ Then star must contract

WHAT IS A STAR?

- ▶ Two conditions satisfied
 - ▶ Bound by self-gravity=
 - ▶ Radiates energy supported by internal source
 - ▶ Nuclear energy → finite resource
 - ▶ Gravitational potential energy → also finite
- ▶ Therefore, stars must 'evolve'
- ▶ Properties vary with time
 - ▶ Density: $\rho(m, t)$,
 - ▶ Temperature: $T(m, t)$
 - ▶ mass fractions, $X_i(m, t)$
 - ▶ N composition elements in star ($1 < i < n$)

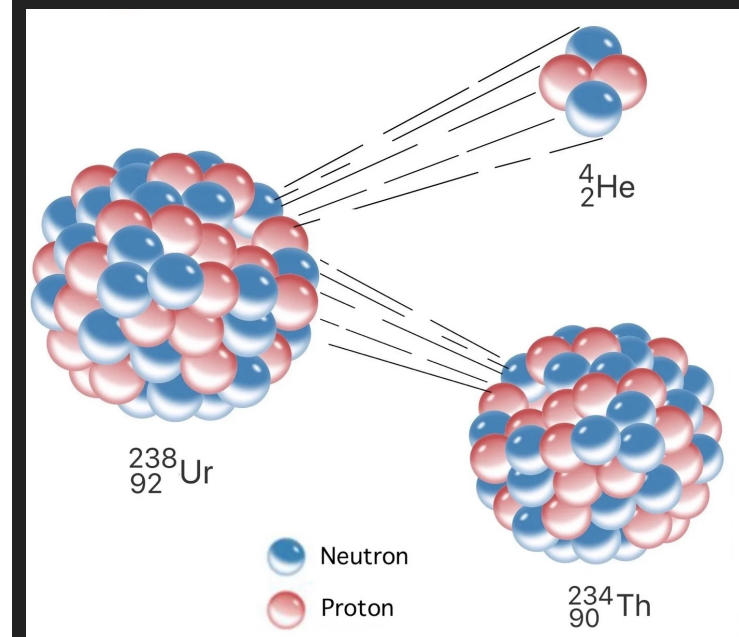
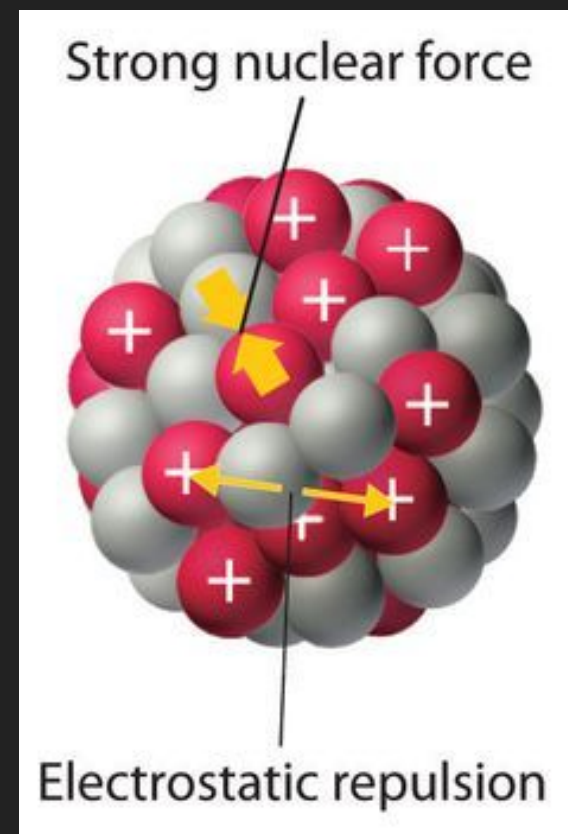
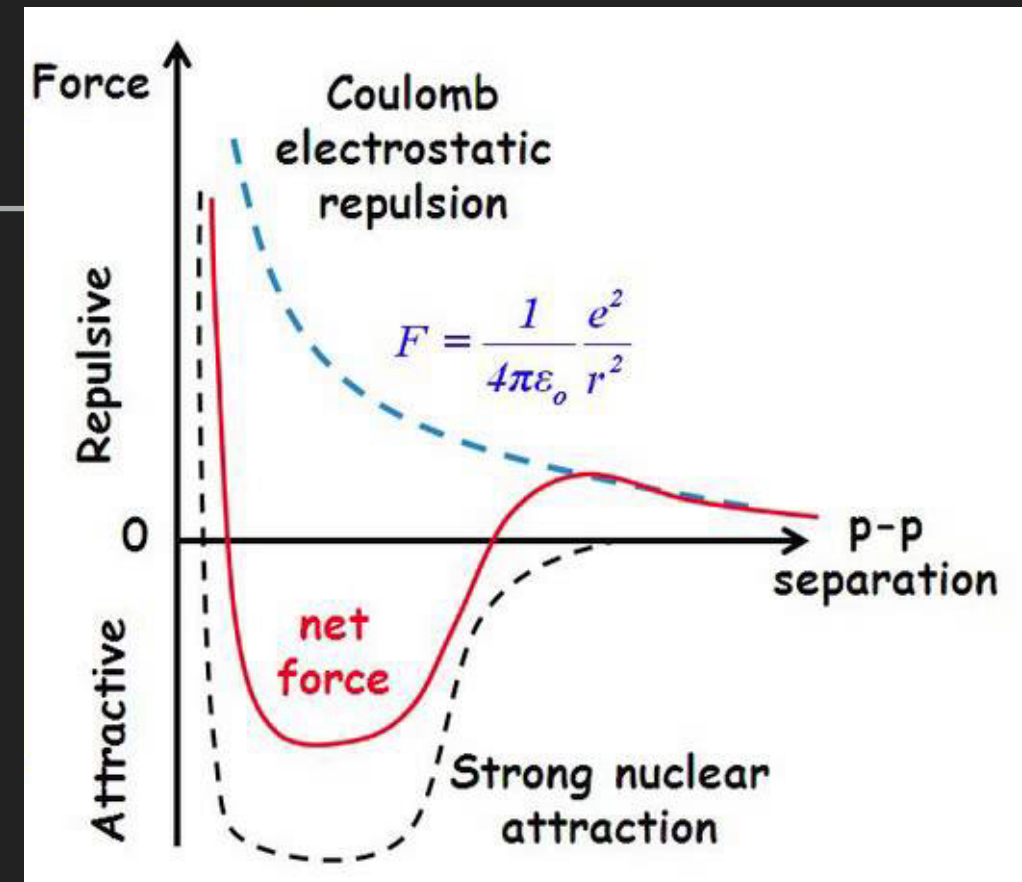
COMPOSITION CHANGES AND ATOMS

- ▶ Plasma: changes not chemical: Nuclear reactions
- ▶ Atomic nucleus
 - ▶ Protons, neutrons (nucleons)
 - ▶ Very similar masses
 - ▶ Baryon # (B or A) and charge (Q or Z)
- ▶ Electrons
 - ▶ Charge, but no baryon #
 - ▶ Very light ($1/2000 m_p$), Lepton #
 - ▶ Partners are neutrinos – zero baryon # and charge
 - ▶ Nearly zero mass
- ▶ Antiparticles
 - ▶ For all baryons and leptons
 - ▶ Baryon and charge reverse signs



STRONG NUCLEAR FORCE

- ▶ A residual force
 - ▶ from fundamental strong interaction
 - ▶ Bonds quarks together to make baryons
 - ▶ Binds protons and neutrons in nucleus
 - ▶ Analogous to chemical bonds of neutral atoms
- ▶ Leptons do not feel this force
- ▶ Short range. $R \leq 10^{-14}\text{m}$
 - ▶ Stronger than Coulomb force involved in p-p repulsion
 - ▶ Distance felt roughly size of largest nuclei
- ▶ Results in alpha and gamma decay

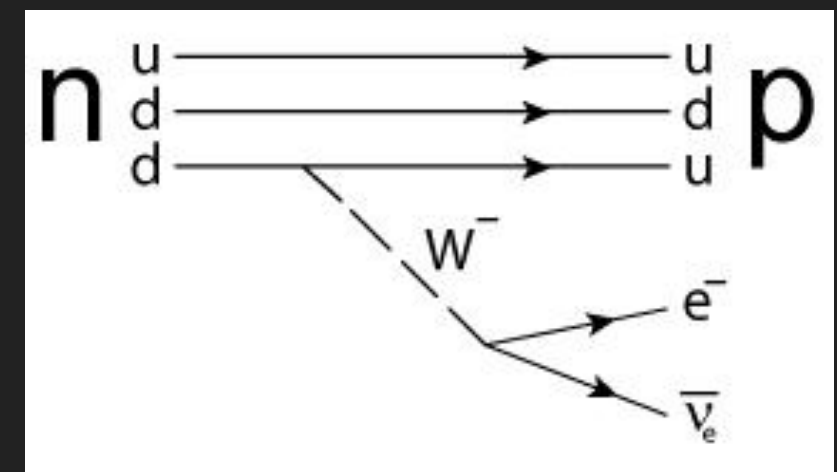
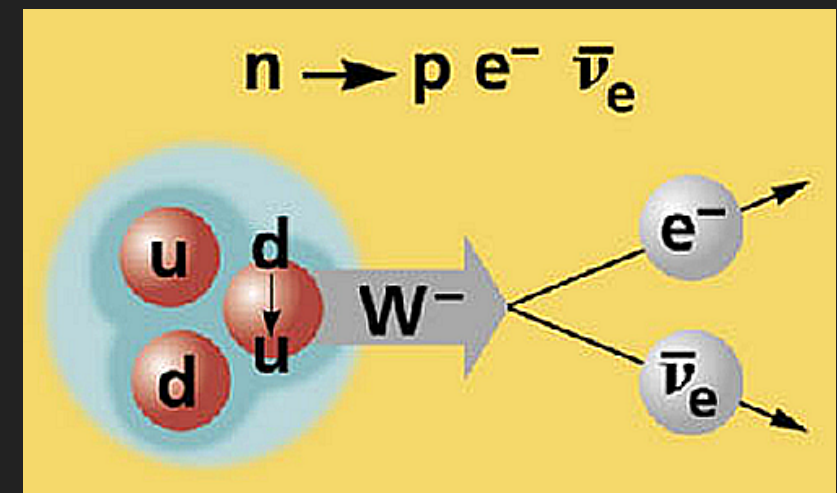


WEAK AND STRONG NUCLEAR FORCES

- ▶ Weak interactions
 - ▶ Also short range (less than strong interaction)
 - ▶ Results in beta decay
 - ▶ Converts proton and neutron to each other
 - ▶ Depends on reaction environment
- ▶ Nuclear forces
 - ▶ Conserve baryon #, lepton # and charge
 - ▶ A, L, Z when dealing w/nuclei
 - ▶ B, L, Q, respectively, in particle physics

Quark structure of nucleons:

- $udd=n$
- $uud=p$



Property	Initial	Final
Q	0	$+1-1+0=0$
L	0	$0+1-1=0$
B	1	$1+0+0=1$

MASS FRACTION OF SPECIES

- ▶ Mass fraction of species, $X_i = \frac{\rho_i}{\rho}$
- ▶ Number density for species, i , $n_i = \frac{\rho_i}{A_i m_H}$
 - ▶ A_i and ρ_i are **baryon #** and **partial density of species**, respectively
 - ▶ m_H is **atomic mass unit** ($\equiv \frac{1}{12} m_{C^{12}}$)
 - ▶ Not quite mass of nucleons
 - ▶ Due to binding energy
 - ▶ Also, protons and neutrons each slightly different mass
- ▶ As function of overall density, ρ

$$n_i = \frac{\rho X_i}{m_H A_i}, \quad X_i = n_i \frac{A_i}{\rho} m_H \quad (\text{P2.47})$$

REACTIONS

- ▶ Changes to # of species 'i'
 - ▶ Due to nuclear fusion or fission
- ▶ Reactions generically



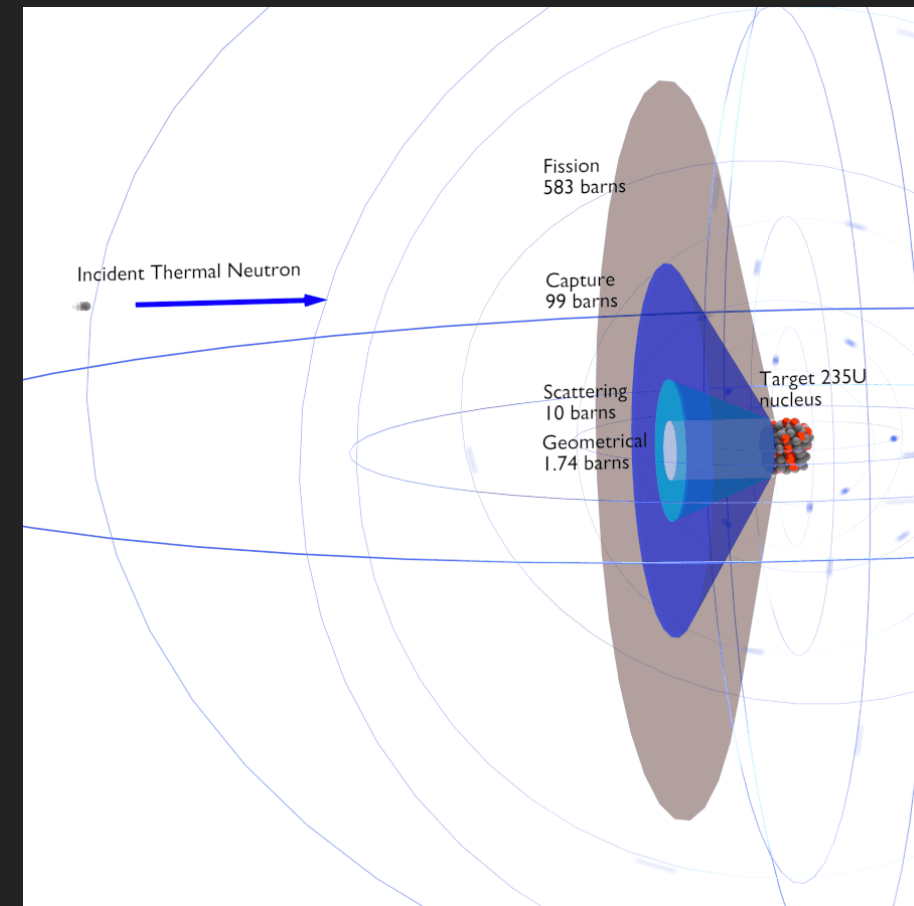
- ▶ Go in either direction
 - ▶ Depends on T or KE and density of particles
- ▶ Constraints $A_i + A_j = A_k + A_\ell$ (P2.49) Baryon # conservation

$$Z_i + Z_j = Z_k + Z_\ell. \quad (\text{P2.50}) \quad \text{Charge conservation}$$

$$L_i + L_j = L_k + L_\ell \quad \text{Lepton \# conservation}$$

NUCLEAR REACTION RATES

- ▶ Three of four particles in reaction determine fourth
 - ▶ So label 2 in and one out: R_{ijk} , or R_{lki}
- ▶ Consider each nucleus has an effective size or 'cross section', σ
 - ▶ Sets probability of interaction
- ▶ Consider nucleus I at rest, and J moving at velocity ' v '
 - ▶ **Result of thermal energy**



$$= n_i n_j \sigma v = n_i n_j R_{ijk}$$

$$\text{▶ Note: if } I = J, \text{ get } = \frac{1}{2} n_i^2 R_{iik}$$

R_{ijk} is 'reaction rate' = σv

$$X_i = n_i \frac{A_i}{\rho} m_H \quad \text{P(2.47)}$$

RATE OF CHANGE OF I'TH SPECIES

- ▶ From all possible nuclear reactions

$$\dot{n}_i = -n_i \sum_{jk} \frac{n_j}{1 + \delta_{ij}} R_{ijk} + \sum_{\ell k} \frac{n_\ell n_k}{1 + \delta_{\ell k}} R_{\ell ki}$$

- ▶ Rate of change of ith element's abundance
 - ▶ First term removes ith element: rate i interacts with j to give k
 - ▶ 2nd term adds it: rate k interacts with l to give i
- ▶ Converting to **rate of change of mass fraction** expression

$$\frac{\dot{X}_i}{A_i} = \frac{\rho}{m_H} \left[\frac{X_i}{A_i} \sum_{jk} \frac{X_j}{A_j} \frac{R_{ijk}}{1 + \delta_{ij}} + \sum_{\ell k} \frac{X_\ell X_k}{A_\ell A_k} \left(\frac{R_{\ell ki}}{1 + \delta_{\ell k}} \right) \right] \quad \text{(P2.52)}$$

- ▶ Can be represented vectorially

- ▶ # of elements is number of species, $\dot{\vec{X}} = \vec{f}(\rho, T, \vec{X}) \quad \text{(P2.53)}$

EVOLUTION

► Primary conservation laws needed:

► Conservation of energy

► Energy equation,
$$\dot{U} + P \frac{\partial}{\partial t} \left(\frac{1}{\rho} \right) = q - \frac{\partial F}{\partial m} \quad (\text{P2.6})$$

► Conservation of momentum

► Equation of motion,
$$\ddot{r} = - \frac{Gm}{r^2} - 4\pi r^2 \frac{\partial P}{\partial m}. \quad (\text{P2.13})$$

► Rate of change of species' abundances

► Rate of change of mass fractions,
$$\dot{\vec{X}} = \vec{f}(\rho, T, \vec{X}). \quad (\text{P2.53})$$

► Thermal changes

► Dynamical/structural changes

► Composition changes

EVOLUTION EQUATIONS

- ▶ Need to know
 - ▶ Structure functions, $\rho(m, t); T(m, t); X_i = (m, t)$
 - ▶ Several areas of physics:
 - ▶ P and U - thermodynamics and statistical mechanics;
 - ▶ F - atomic physics and radiation transfer;
 - ▶ q and f - nuclear and particle physics
 - ▶ Boundary conditions for
 - ▶ 2 space derivatives
 - ▶ N+3 time derivatives
- ▶ A tall order – need further work!
 - ▶ what are the timescales processes occur?

EVOLUTION EQUATIONS

- ▶ TE and HE,

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4}$$

$$\frac{dF}{dm} = q$$

$$\dot{\vec{X}} = \vec{f}(\rho, T, \vec{X})$$

- ▶ Much simpler equations
 - ▶ Don't need to know all initial structure
 - ▶ Need initial composition